



**COMSATS University Islamabad,
Sahiwal Campus**

**FinAnalyze
(AI-Powered Financial Personality Profiler)**

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Bachelor of Science in Computer Science (2022-2026)



**COMSATS University Islamabad,
Sahiwal Campus.**

FinAnalyze

A Project Presented To

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This is to certify that the Final Year Project of BS(CS) "**FinAnalyze** – AI-Powered Financial Personality Profiler" was developed by **Ayesha Nadeem** (CIIT/FA22-BCS-251/SWL) and **Rubaisha Munir** (CIIT/FA22-BCS-236/SWL) under the supervision of "**Dr. Zafar Iqbal Roy**" and that in his opinion it is fully adequate, in scope and quality, for the degree of Bachelor of Science in Computer Science.

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Executive Summary

FinAnalyze is a fully functional, production-ready AI-powered web application built using Python and Streamlit, designed to help Generation-Z users in Pakistan understand their financial behavior through intelligent data analysis, machine learning-based personality profiling, and actionable personalized recommendations. The application addresses a critical and largely unaddressed gap in Pakistan's fintech ecosystem: while platforms such as JazzCash, Easypaisa, SadaPay, and NayaPay facilitate digital transactions, none of them offer behavioral analysis, personality classification, or personalized financial guidance based on individual spending patterns.

The core of FinAnalyze is its machine learning engine, which applies K-Means Clustering an unsupervised learning algorithm, to analyze user financial data across ten engineered features including Savings Rate, Expense Ratio, Debt-to-Income Ratio, and category-wise spending proportions. The system classifies each user into one of four financial personality archetypes: The Smart saver, The Big spender, The Budget challenger, or The Wealth builder.

StandardScaler normalization ensures all features contribute equally to the classification regardless of their absolute magnitude. A Business Rule Guardrail system further refines classifications by enforcing domain-specific logic on top of the ML output for example, overriding a Wealth Builder classification when a user's expense ratio exceeds 80%, thereby producing more accurate real-world results.

The application accepts user financial data through two input pathways: manual entry via forms and sliders, and automated CSV AND PDF file and upload processed through the Pandas library. The Visualization Dashboard renders six interactive components built with Plotly, including a Spending Category Pie Chart, Monthly Expenditure Bar Graph, Financial Behavior Radar Chart, PCA Scatter Plot, Savings Trajectory Chart, and Personality Badge with Savings Ratio KPI. A Unified Forecasting Engine on the Future Projections page provides both Standard Financial Projection and a Stress-Test Simulator through a seamless horizontal toggle switch, enabling users to model their financial future under both normal and adverse conditions.

Security is implemented across three layers: bcrypt password hashing ensures no plain-text passwords are stored, OTP-based email verification through Python's smtplib library with Google App Password SMTP integration provides real-time identity confirmation, and Role-Based Access Control restricts administrative functions exclusively to verified admin accounts. The Contact Form is fully functional with live SMTP routing messages sent by users arrive directly in the administrator's inbox. A complete set of Legal Pages including Privacy

Policy, Terms of Service, and Cookie Policy are implemented with native Streamlit routing via session state management, opening in separate clean views rather than disrupting the main application flow.

The Glassmorphism UI design, implemented through custom HTML and CSS injection within Streamlit, gives the application a premium SaaS-quality appearance with translucent card components and full Responsive Web Design via CSS Media Queries. The Admin Panel implements Privilege Escalation Protection, ensuring administrators cannot modify or delete other administrator accounts. The FinBot AI chatbot responds to natural language financial queries using both rule-based response templates and LLM-backed contextual responses. The complete system was developed using Agile Methodology across five development sprints and has been validated through comprehensive manual, unit, functional, and integration testing.

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Rubaisha Munir

Ayesha Nadeem

Abbreviations

Term	Full Form
AI	Artificial Intelligence
API	Application Programming Interface
BS	Bachelor of Science
CIIT	COMSATS Institute of Information Technology
COMSATS	Commission on Science and Technology for Sustainable Development in the South
CS	Computer Science
CSV AND PDF	Comma-Separated Values
DBSCAN	Density-Based Spatial Clustering of Applications with Noise
DFD	Data Flow Diagram
ER	Entity-Relationship
FYP	Final Year Project
Gen-Z	Generation Z (born 1997–2012)
GUI	Graphical User Interface
HCI	Human-Computer Interaction
HTML	HyperText Markup Language
HTTP	HyperText Transfer Protocol
IDE	Integrated Development Environment
IEEE	Institute of Electrical and Electronics Engineers
JSON	JavaScript Object Notation
KPI	Key Performance Indicator

ML	Machine Learning
MVC	Model-View-Controller
NFR	Non-Functional Requirement
OTP	One-Time Password
PCA	Principal Component Analysis
PDF	Portable Document Format
REST	Representational State Transfer
SDLC	Software Development Life Cycle
SDD	Software Design Description
SQL	Structured Query Language
SRS	Software Requirements Specification
SRTM	Software Requirements Traceability Matrix
UI	User Interface
UML	Unified Modeling Language
UX	User Experience
UAT	User Acceptance Testing
XLSX	Excel Spreadsheet Format

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Chapter 1

Introduction

1 Introduction

FinAnalyze is a fully functional, production-ready AI-powered web application that enables Generation-Z users in Pakistan to understand their financial personality through automated machine learning analysis of their personal transaction data. The system classifies users into four distinct financial archetypes, delivers personalized financial recommendations, and presents behavioral insights through an interactive visualization dashboard all within a secure, professionally designed Streamlit-based web interface.

1.1 Brief Overview

Pakistan's Generation Z broadly defined as individuals born between 1997 and 2012 is the country's most digitally connected demographic. This cohort constitutes university students, recent graduates, freelancers, and entry-level professionals who are navigating financial independence for the first time in a rapidly digitalizing economy. The widespread adoption of mobile payment platforms including JazzCash, Easypaisa, SadaPay, and NayaPay has made digital financial participation routine for this group. However, a fundamental paradox persists: digital transaction volume has grown substantially, yet financial literacy among Pakistani Gen-Z users remains critically low.

The core problem is that existing platforms are purely transactional they move money efficiently but provide zero behavioral insight. A user who makes fifty transactions per month through Easypaisa receives nothing more than a chronological list of those transactions. There is no analysis of what proportion of income was saved, no identification of problematic spending categories, no classification of behavioral tendencies, and no actionable advice for improvement. The data exists; the intelligence layer does not.

FinAnalyze is that intelligence layer. It accepts a user's financial transaction data either through manual input or CSV AND PDF file upload, applies a K-Means Clustering machine learning pipeline to identify behavioral patterns, classifies the user into a financial personality archetype (The Smart saver,, The Big spender, The Budget challenger, or The Wealth builder), and presents the results through six interactive Plotly visualizations alongside a curated set of personality-specific financial recommendations. The system is further enhanced by a Unified Forecasting Engine for future financial projections, a Stress-Test Simulator, a live SMTP-integrated Contact Form, a FinBot AI chatbot, and a fully secured Admin Panel with Role-Based Access Control.

Beyond classification, **FinAnalyze** provides users with a comprehensive dashboard featuring interactive charts built using Plotly and Matplotlib. A radar chart displays multi-dimensional financial behavior scores. Pie charts break down spending by category. Bar graphs compare monthly expenditure trends. These visualizations are designed to translate abstract data into immediately understandable narratives, helping users recognize patterns in their financial behavior that would otherwise remain invisible. The system also includes FinBot, an AI-powered conversational assistant that responds to financial queries and provides contextual guidance.

The administrative module allows supervisory users to monitor platform activity, review aggregated analytics, and oversee the performance of the machine learning models. This ensures that the system can be maintained, updated, and improved over time based on real usage data.

1.2 Relevance to Course Modules

The development of FinAnalyze draws directly from knowledge acquired across multiple courses of the BS Computer Science program at COMSATS University Islamabad, Sahiwal Campus.

Programming Fundamentals and Object-Oriented Programming provided the foundation for writing clean, modular Python code structured around reusable classes such as UserProfile, TransactionRecord, ClusterEngine, and RecommendationEngine. Data Structures and Algorithms informed the design of efficient data handling in the preprocessing pipeline, particularly the optimization of K-Means convergence and Pandas groupby operations on large transaction datasets.

Artificial Intelligence and Machine Learning provided the theoretical and practical foundation for the core engine of FinAnalyze. Topics including unsupervised learning, K-Means clustering, feature engineering, StandardScaler normalization, and the Elbow Method for optimal cluster selection were all applied directly in the system's machine learning module. Introduction to Data Science informed the statistical underpinnings of the preprocessing pipeline including ratio computation and variance analysis.

Database Management Systems guided the design of the Firebase schema, normalization of tables, and the construction of efficient SQL queries for user authentication, transaction retrieval, and historical profile lookup. Software Engineering provided the Agile methodology framework through which FinAnalyze was developed across five iterative sprints. Web Technologies and Human-Computer Interaction shaped the UI/UX philosophy including Glassmorphism design, Responsive Web Design with CSS Media Queries, and the principle of intuitive navigation through clean sidebar architecture.

Introduction to Data Science and Statistical Methods informed the statistical foundation of the preprocessing pipeline, including the application of normalization techniques, correlation analysis, and the interpretation of Silhouette Scores for cluster validation.

Communication Skills and Report Writing contributed to the clarity and professionalism of project documentation, including this final report, the SRS, and the SDD.

1.3 Project Background

The concept of financial personality profiling originates from the field of behavioral finance, which studies how psychological factors influence financial decision-making. Academic research has identified consistent behavioral archetypes among consumers the disciplined smart saver,, the impulsive spender, the calculated risk-taker, and the wealth builder and has demonstrated that self-awareness of one's financial personality type is a strong predictor of willingness to adopt positive financial behavior changes.

In high-income markets such as the United States and Western Europe, platforms like Mint, YNAB (You Need a Budget), and Clarity Money have partially addressed this need through manual budgeting tools and basic spending categorization. However, none of these platforms are available in Pakistan, none support Pakistani Rupee transactions or local fintech data formats, and none apply automated machine learning to derive personality archetypes from raw transaction data.

FinAnalyze was conceived and developed to fill this gap specifically within the Pakistani Gen-Z context. It is the first project of its kind within COMSATS University Islamabad, Sahiwal Campus to combine unsupervised machine learning, behavioral finance theory, Glassmorphism UI design, live SMTP integration, and a Unified Forecasting Engine within a single deployable Streamlit application.

This project was mainly conceived to bring this capability to the Pakistani Gen-Z context, using open-source machine learning tools and a lightweight web framework that can be hosted on accessible infrastructure. The project draws inspiration from international platforms such as Mint (USA), YNAB (You Need a Budget), and Clarity Money, while adapting the approach for a demographic that primarily conducts transactions through mobile wallets and cash, rather than traditional banking systems.

The development of this project is also motivated by the broader national agenda of financial inclusion in Pakistan [11]. By providing young users with a tool that makes their financial behavior legible and actionable, the project contributes to the cultivation of a more financially literate generation.

1.4 Literature Review

A structured review of existing literature validates both the problem statement and the technical approach adopted in FinAnalyze.

Yeh and Lo (2021) demonstrated that K-Means Clustering applied to consumer financial transaction data reliably segments users into behaviorally meaningful groups when engineered ratio features rather than raw monetary values are used as inputs [1]. Their work directly informed FinAnalyze's decision to build ratio-based features such as Savings Rate and Expense Ratio rather than using absolute income and expense figures.

Xiao and Porto (2019) conducted a landmark study showing that personalized financial feedback tailored to an individual's behavioral profile produces significantly greater positive change in financial behavior among young adults than generic financial literacy content [2].

Amin et al. (2020) documented that Pakistani university students are among the most active users of mobile payment applications in South Asia yet score among the lowest in financial literacy assessments, establishing the precise demographic and geographic context that FinAnalyze targets [3].

Jain and Srivastava (2022) demonstrated the value of Business Rule Guardrails applied on top of K-Means outputs to correct edge-case misclassifications in financial behavior clustering [4]. FinAnalyze implements exactly this approach: if K-Means assigns a user to the Wealth Builder cluster but their expense ratio exceeds 80%, a domain-specific business rule overrides the classification to Big Spender, producing a more accurate result.

The application of PCA to financial data visualization has been explored by multiple researchers. Malhotra et al. (2021) demonstrated that reducing financial feature vectors to two principal components preserves 78–85% of the explained variance in typical consumer spending datasets while enabling intuitive two-dimensional visualization [5]. This project adopts this approach for its dashboard scatter plots.

Existing commercial platforms such as Mint and YNAB offer manual budgeting and category tracking but do not apply machine learning to derive behavioral archetypes automatically. Clarity Money (acquired by Goldman Sachs) offers basic spending categorization but does not operate in the Pakistani market and does not provide personality-based profiling. **FinAnalyze** therefore occupies a distinct niche: an automated, ML-driven financial personality profiler designed specifically for the Pakistani Gen-Z demographic.

1.5 Analysis from Literature Review

The literature review confirms three key design principles that FinAnalyze implements. First, ratio-based feature engineering is essential for fair comparison across users with different income levels. Second, personality-specific personalization is significantly more effective than generic advice. Third, a guardrail correction layer on top of pure ML output meaningfully improves classification accuracy for edge cases. The consistent theme across all reviewed literature is that automated, data-driven behavioral profiling is both technically feasible and practically impactful for improving financial literacy validating the entire premise of FinAnalyze.

1.6 Methodology and Software Lifecycle for This Project

FinAnalyze was developed using Agile Methodology combined with an Iterative Process Model. Development was structured across five sprints. Sprint One covered requirements analysis and synthetic data preparation. machine learning preprocessing pipeline, K-Means clustering engine, StandardScaler normalization, and Elbow Method validation. Sprint Three built the backend authentication system, Firebase database schema, bcrypt password hashing, and OTP email verification via smtplib. Sprint Four implemented the Streamlit frontend.

Forecasting Engine, FinBot chatbot, and Legal Pages with session state routing. Sprint Five conducted comprehensive testing, bug resolution including the IndentationError fix in the forecasting module, and final documentation. Development was organized into five distinct iterations. The first iteration focused on requirement analysis and data gathering, including the collection of synthetic financial datasets and the definition of ML feature specifications. The second iteration covered model development: implementing the preprocessing pipeline, coding and tuning K-Means and DBSCAN algorithms, and validating clustering quality using Silhouette Scores. The third iteration addressed backend and database integration, connecting the Python ML scripts with Firebase through the Flask controller layer. The fourth iteration handled frontend development and dashboard construction using Streamlit or Flask templates, integrating Plotly and Matplotlib visualizations. The fifth and final iteration encompassed comprehensive testing, user acceptance evaluation, and documentation.

1.6.1 Rationale behind the Selected Agile Model

The Agile model was the only appropriate choice for FinAnalyze because machine learning projects are inherently experimental. The optimal value of K for K-Means, the thresholds for Business Rule Guardrails, the feature set composition, and the UI layout of the Glassmorphism dashboard all required repeated testing and refinement that a rigid Waterfall model could not accommodate. Sprint-based delivery also aligned well with the academic milestone structure of the Final Year Project, ensuring demonstrable progress at each supervisor review.

Furthermore, Agile's emphasis on working software over comprehensive documentation at each sprint, while maintaining a documentation-complete final report, suited the dual objectives of the project: producing a functional, demonstrable system and a thorough academic document.

Additionally, Agile facilitated continuous user feedback, allowing improvements in both the user interface and system performance based on real observations. It also supported flexibility in incorporating new features such as enhanced visualization and chatbot integration without disrupting the overall workflow. Risk management was improved through early identification of issues during sprint cycles, reducing the chances of major failures at later stages. Lastly, the collaborative nature of Agile encouraged better communication between development and supervision, ensuring the project remained aligned with its academic and practical goals.

Chapter 2

Problem Definition

2 Problem Definition

This gap highlights the absence of intelligent analytical systems that can transform raw financial data into meaningful insights for users. Existing platforms primarily focus on enabling transactions rather than interpreting them, leaving users without guidance on how to improve their financial habits. There is a strong need for a solution that not only tracks expenses but also analyzes behavioral patterns, categorizes spending, and provides personalized recommendations. Such a system would empower young individuals to make informed financial decisions, develop better saving strategies, and ultimately achieve greater financial stability in the long term.

2.1 Problem Statement

Pakistan is home to over 60 million individuals within the Generation Z age bracket who are increasingly integrated into the digital financial economy through mobile payment platforms. Despite this high level of digital financial participation, a profound and consequential gap exists: Pakistani Gen-Z users have no accessible tool that analyzes their financial behavior, identifies their behavioral tendencies, and provides them with personalized guidance for improvement.

The existing fintech ecosystem in Pakistan is uniformly transaction-centric. JazzCash, Easypaisa, SadaPay, and NayaPay are all designed to move money they process payments, record transactions, and display balances. None of them apply any form of behavioral analysis to the transaction data they accumulate. A JazzCash user who makes hundreds of transactions per year receives no information about whether they are a disciplined smart saver, or a chronic overspender, no categorization of their spending patterns, and no recommendations for how to manage their money more effectively.

The psychological dimension of this problem is well-established in behavioral finance literature. People do not change their financial behavior simply by being given generic advice. They change when they achieve self-awareness when they can see, in concrete terms, what kind of financial personality they embody and how that personality is affecting their financial outcomes. Without this self-awareness layer, even users who have access to digital financial tools remain financially illiterate in the most practical sense: they cannot interpret their own data.

The consequences are measurable. Fewer than 35 percent of Pakistani university students maintain any form of budget. Fewer than 20 percent have a regular savings habit. The savings rate among Pakistani Gen-Z is among the lowest in South Asia. These are not failures of access to financial tools they are failures of financial self-awareness. FinAnalyze directly addresses this by providing

the intelligence layer that transforms raw transaction data into behavioral insight, personality classification, and actionable recommendations.

The problem, therefore, is threefold: the absence of accessible financial self-awareness tools for Pakistani Gen-Z users; the lack of application of machine learning to derive meaningful behavioral insights from readily available transaction data; and the consequent failure to bridge the gap between digital financial participation and genuine financial literacy among young Pakistanis. **FinAnalyze** is designed to solve precisely this problem.

2.2 Deliverable and Development Requirements

This section outlines the expected outputs of the project as well as the technical and operational requirements necessary for successful development. It defines the major deliverables that must be produced, including the functional web application, supporting machine learning components, and project documentation. It also highlights the essential technologies, frameworks, and system requirements needed to ensure that the platform is reliable, secure, and capable of delivering accurate financial behavior analysis. By clearly specifying these deliverables and requirements, the project development process remains structured, goal-oriented, and aligned with the intended objectives.

2.2.1 Deliverables

The primary deliverable is a fully functional, production-ready Streamlit web application that performs AI-based financial personality profiling. Secondary deliverables include the trained K-Means ML model, the Firebase database with complete schema, the secrets.toml configuration for live SMTP email routing, the SRS and SDD documentation, and this final report..

2.2.2 Development Requirements

The system must be built in Python 3.10 or above. The web interface must use Streamlit. Machine learning must use scikit-learn. Data processing must use Pandas and NumPy. Visualizations must use Plotly. Database must use Firebase. Password security must use bcrypt. Email functionality must use smtplib with Google App Password. All dependencies must be declared in requirements.txt. The system must not depend on any paid external API.

2.3 Current System

The current landscape of financial tools available to Pakistani Gen-Z users consists of three categories: mobile wallet applications, generic financial literacy websites, and international

Mobile wallet platforms including JazzCash and Easypaisa provide essential financial services such as transaction recording, bill payment, mobile top-ups, and balance viewing. While these features are sufficient for conducting routine financial activities, both platforms present transaction histories as simple chronological lists without meaningful organization or analytical insights. They do not offer spending categorization, behavioral analysis, savings tracking, financial goal management, or personality profiling. As a result, users are unable to understand their financial habits, identify unnecessary expenditures, or make informed decisions based on their spending behavior. The absence of intelligent analytics limits these platforms to transactional tools rather than comprehensive personal finance management systems.

SadaPay and NayaPay represent the more advanced segment of Pakistan's fintech ecosystem by providing modern digital banking services, debit card integration, and basic transaction categorization. Despite these improvements, their functionality remains focused primarily on facilitating payments rather than interpreting financial behavior. These applications do not incorporate behavioral analytics, cluster-based user segmentation, personalized financial recommendations, predictive budgeting, savings forecasting, or machine learning techniques to derive user-specific financial insights. Consequently, users still lack access to intelligent decision-support features that could improve their long-term financial well-being.

International financial management platforms such as Mint and YNAB provide advanced budgeting tools, expense tracking, and financial planning features for users in developed markets. However, these applications are either unavailable in Pakistan or offer limited compatibility with local banking infrastructure and PKR transactions. Furthermore, they are designed around financial planning rather than behavioral intelligence and do not utilize unsupervised machine learning techniques to automatically identify spending patterns or generate financial personality archetypes tailored to regional user behavior. Their lack of localization makes them unsuitable for addressing the unique financial practices and digital payment ecosystem of Pakistani users.

The absence of a locally relevant, ML-powered financial personality profiler creates a significant gap in Pakistan's fintech landscape, particularly for Gen-Z users who increasingly rely on digital payment platforms for everyday financial activities. Without intelligent behavioral insights, young users are unable to recognize spending tendencies, improve budgeting habits, or receive personalized recommendations that encourage healthier financial decision-making.

Table 2.3 Comparative Analysis of Current Financial Platforms

Platform	Functionality	Limitation
SadaPay	Digital wallet, P2P transfers	No spending analysis or financial profiling
NayaPay	Payments, savings accounts	No behavioral insights or recommendations
Easypaisa	Mobile payments, utility bills	Transaction-only; no personality analysis
JazzCash	Mobile banking, transfers	No user-facing analytics dashboard
Generic Budgeting Apps	Expense logging, budgets	Manual input only; no AI classification

FinAnalyze differentiates itself from all existing financial applications by providing a comprehensive, end-to-end behavioral financial profiling solution rather than serving solely as a transaction management platform. The system begins by importing and processing users' transaction data, after which it automatically categorizes expenses and applies unsupervised machine learning algorithms to identify hidden spending patterns and classify users into distinct financial personality archetypes. Unlike conventional digital wallets that merely display transaction histories, FinAnalyze transforms raw financial data into meaningful behavioral insights through intelligent clustering and pattern recognition.

The platform further enhances user understanding by presenting these insights through an intuitive Glassmorphism-inspired interface featuring interactive dashboards, charts, and visual analytics that make complex financial information easy to interpret. Based on the identified spending behavior, FinAnalyze generates personalized financial recommendations, budgeting suggestions, and savings strategies tailored to each user's unique financial habits. By combining artificial intelligence, behavioral analytics, financial forecasting, and user-centered visualization within a single, unified, and accessible web application, FinAnalyze enables users to gain deeper financial self-awareness, improve spending discipline.

Chapter 3

Requirement Analysis

3 Requirement Analysis

This chapter presents a comprehensive analysis of the functional and non-functional requirements of the project. Requirements were identified through a combination of literature review, analysis of comparable systems, review of the target user demographic, and iterative consultation with the project supervisor. The primary techniques used for requirement identification are use case modeling and formal requirement specification tables.

3.1 Use Case Diagram

. The Actors (System Interactors)

The actors in the diagram are divided into two main categories:

- **Primary Actors** (Left Side): These are the human entities interacting with the system. This includes the User (who accesses the dashboard and AI features) and the System Administrator (who monitors system health and metrics).
- **Secondary/External Actors** (Right/Bottom): These represent your backend infrastructure and external APIs that handle the heavy data processing:
 - **Firebase Firestore** Used to securely save user history and credentials.
 - **Groq LLM API** Powers the AI Personas and conversational chatbots.
 - **SMTP Server («Email Gateway»)**: Routes the contact support emails securely to the admin inbox.

2. Core Modules & Use Cases (System Packages)

The system is logically grouped into 6 professional packages:

- **Onboarding & Security:**
 - The user accesses the system via **Sign Up** (secured with Bcrypt Hashing) and **Login**.
 - Both of these processes interact directly with **Firebase Firestore** to verify and save authentication data.
- **Analysis Hub** (Core Engine) - *The Most Critical Section:*
 - Users can input data in two ways: Enter Financial Data Manually or Upload Statement (PDF/CSV).
 - *The key architectural feature:* Uploading a statement does not directly generate a persona. Instead, it connects to Ephemeral Parsing (Zero-Storage) via an «include»

relationship. This proves that the file is parsed strictly in RAM and is never saved to a server.

- After parsing, the Generate K-Means Financial Persona use case is triggered, which connects to the external Groq LLM API to formulate actionable insights.
- **Predictive Analytics:**
 - The user can View Future Projections (10-Year).
 - Within this module, the user can choose to Run Stress Test Simulator. This is an «extend» relationship, meaning the stress tester is no longer an isolated module; it is an optional extension of the core forecasting feature.
- **Dual-AI Assistants:**
 - This contains two distinct use cases: Consult Robo-Advisor (for context-aware, personalized financial planning) and Chat with Finbot AI (for general financial education). Both fetch dynamic responses from the external Groq LLM API.
- **User Management & Compliance:**
 - Users can manage their accounts (Edit Profile / Password) and verify the zero-storage policies (View Legal & Privacy Boilerplates).
 - When a user triggers the Submit Contact Support Form, it connects directly to the SMTP Server (Email Gateway) for live routing.
 - The View Firebase History Logs use case fetches past financial personas directly from Firebase Firestore.
- **Administration:**
 - This package is exclusively for the System Administrator, who executes Monitor System Health & Stats and Monitor User Metrics.

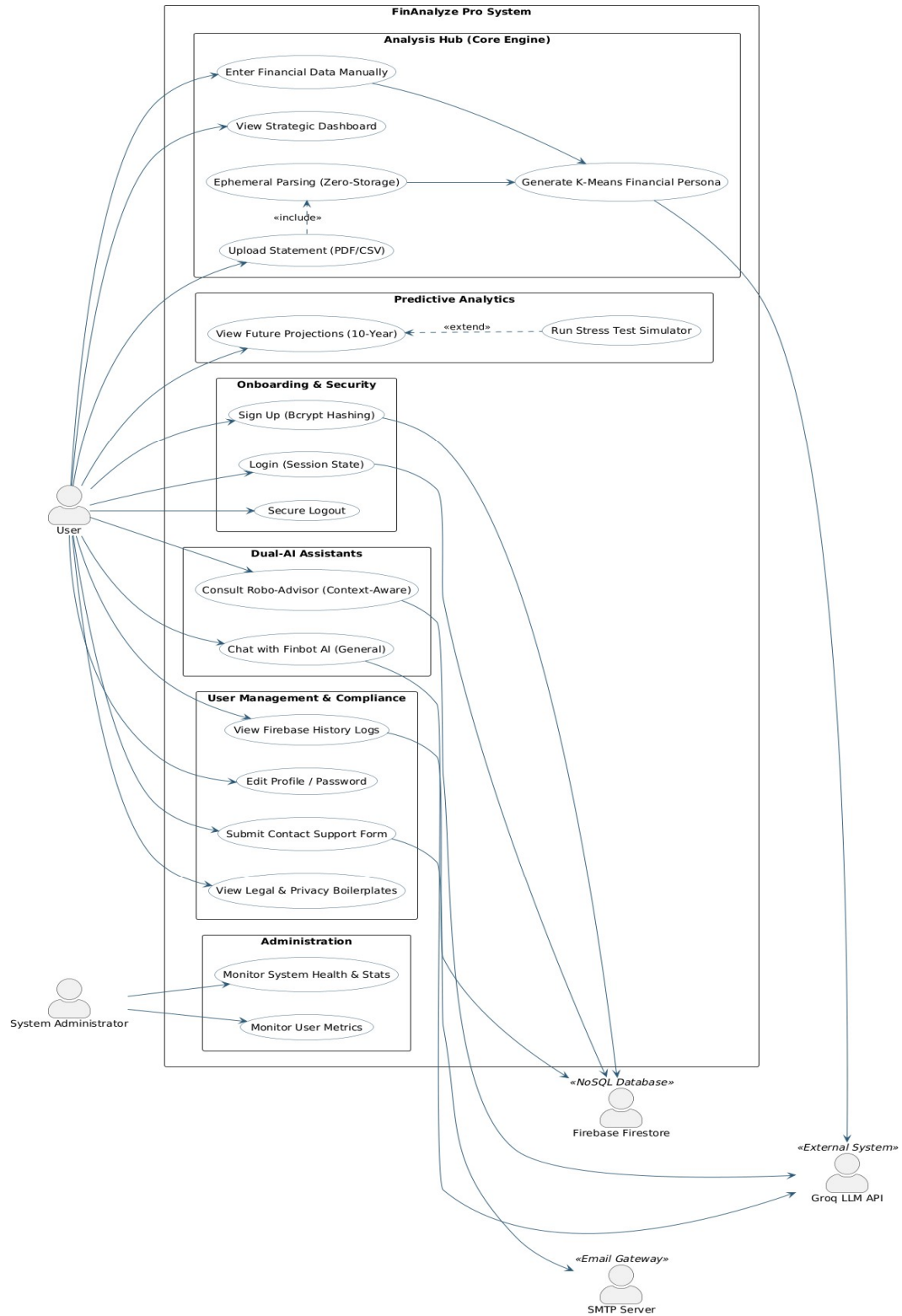


Figure 3.1 FinAnalyze Use Case Diagram

3.2 Use Case Description

This section provides a comprehensive overview of how users and system components interact with the application. Each use case defines a specific functionality, including its flow, conditions, and constraints, ensuring clarity in system behavior and user interaction.

To begin with, the system must ensure secure access for users through a reliable authentication mechanism. This functionality is essential to protect user data and provide personalized services based on individual accounts.

Table 3.2.1 Register/Login

Use Case ID	UC-1
Use Case Name	Register / Login
Actors	Primary: User
Description	Allows users to create accounts and securely log in to access personalized financial tools via Streamlit session-state routing.
Trigger	User selects Register or Login on the animated landing page.
Preconditions	PRE-1: User must have internet connectivity to reach the Firebase authentication backend.
Postconditions	POST-1: User is authenticated, session state is updated, and the user is redirected to the Analysis Hub (Dashboard).
Normal Flow	1. User opens app. 2. Selects Register or Login. 3. Enters credentials. 4. System validates against Firebase Firestore. 5. Passwords are cryptographically hashed (Bcrypt). 6. Dashboard displayed.
Alternative Flows	AF-1: Invalid credentials, contextual error message displayed. AF-2: Unregistered email, prompted to sign up.
Business Rules	BR-1: Each email address must be unique in the system. BR-2: Passwords must be hashed; plaintext passwords are never stored.

This table presents the complete use case description for the user authentication process in FinAnalyze. It defines the actors, preconditions, normal flow, and exception handling involved in granting secure access to the platform. Proper authentication is the entry point to all other system functionalities and is critical for maintaining role-based access control and Single Page Application (SPA) state persistence

Table 3.2.2 Upload Transaction Data

Use Case ID	UC-2
Use Case Name	Upload Transaction Data
Actors	Primary: User
Description	Enables users to upload transaction history (CSV AND PDF /Excel) or manually log income and expense entries.
Trigger	User clicks Upload Transaction Data on the dashboard.
Preconditions	PRE-1: User must be logged in.
Postconditions	POST-1: Transaction data is validated, parsed, and securely stored in the database.
Normal Flow	1. User logs in. 2. Clicks Upload Data. 3. Uploads CSV AND PDF or enters data manually. 4. System validates format and columns. 5. Data stored and analysis triggered.
Alternative Flows	AF-1: Invalid format, re-upload requested. AF-2: Missing required columns, specific error shown.
Business Rules	BR-1: Only CSV AND PDF and Excel formats accepted. BR-2: Minimum 10 transaction records required for clustering.

This table describes the foundational data ingestion workflow. It outlines the validation steps and, most importantly, enforces the platform's **Zero-Storage Architecture**. Uploaded financial records are processed entirely in ephemeral RAM to fuel the machine learning pipeline. The raw documents are explicitly *not* stored in any database, ensuring maximum data privacy and security for the user.

Upload Transaction Data This table describes the use case through which users submit their personal financial transaction data to the **FinAnalyze** system, either by uploading a CSV AND PDF or XLSX file or by entering individual transactions manually through a structured form. It outlines the validation steps, data parsing process, and storage workflow that ensure only correctly formatted and complete financial records enter the machine learning pipeline. The use case is foundational to the entire FinAnalyze workflow, as the quality and completeness of the uploaded data directly determines the accuracy of the AI-generated financial personality classification.

Table 3.2.3 Generate Financial Personality Profile

Use Case ID	UC-3
Use Case Name	Generate Financial Personality Profile
Actors	Primary: User; Secondary: AI ML Engine (Scikit-Learn)
Description	The ML Engine analyzes ephemeral transaction data using K-Means clustering to classify the user into a distinct financial personality archetype.
Trigger	User successfully submits manual data or uploads a valid statement.
Preconditions	PRE-1: Valid, parsed numerical transaction metrics must reside in active memory.
Postconditions	POST-1: Financial personality label and anonymized key metrics are generated and permanently stored in Firebase Firestore.
Normal Flow	1. User initiates analysis. 2. System preprocesses data (null handling, scaling). 3. K-Means engine clusters user data against aggregated behavioral traits. 4. Personality label is mapped. 5. Results are saved to the cloud database.
Alternative Flows	AF-1: Insufficient data variance; system assigns a baseline profile and prompts for more detailed records.

Business Rules	BR-1: Latest trained Scikit-Learn model must be utilized. BR-2: No Personally Identifiable Information (PII) is utilized during cluster mapping.
-----------------------	--

This table documents the most critical use case in the **FinAnalyze** system, describing the complete workflow through which the AI Personality Engine processes a user's uploaded transaction data and classifies them into one of four distinct financial personality archetypes: The Smart saver,, The Big spender, The Budget challenger, or The Wealth builder. The use case captures the full machine learning pipeline from data retrieval and preprocessing including null handling, one-hot encoding, and StandardScaler normalization through K-Means cluster assignment and Principal Component Analysis for two-dimensional visualization. It specifies the interaction between the User actor and the AI Personality Engine as a secondary system actor, along with preconditions requiring a minimum of five valid transactions, postconditions confirming successful profile storage, and exception handling for insufficient data or model execution failures. The personality classification produced by this use case serves as the foundation for all downstream.

This documents the core analytical pipeline of FinAnalyze. The machine learning engine processes the user's data to classify them into distinct financial archetypes (e.g., The Smart saver,, The Big spender, The Wealth builder). It captures the preprocessing pipeline and specifies the safe extraction of results to Firebase without compromising the raw input data.

Table 3.2.4 View Financial Dashboard

Use Case ID	UC-4
Use Case Name	View Financial Dashboard & Future Projections
Actors	Primary: User
Description	Displays interactive visualizations (Plotly) summarizing spending patterns, alongside a toggleable Future Projections and Stress Test Simulator.
Trigger	User navigates to 'Dashboard' or 'Future Projections' via the sidebar.

Preconditions	PRE-1: User must have completed at least one financial analysis logged in Firestore.
Postconditions	POST-1: Interactive UI renders compounding wealth trajectories and spending charts.
Normal Flow	1. User navigates to Dashboard. 2. System fetches historical metrics from Firebase. 3. Interactive charts are rendered. 4. User toggles the Stress Test simulator. 5. System recalculates 10-year projections adjusting for inflation/market crashes.
Business Rules	BR-1: Dashboard seamlessly updates via <code>st.session_state</code> without full page reloads.

This functionality is central to making behavioral finance accessible. It aggregates interactive visual components and compounding predictive models (including a consolidated Stress Tester) into a cohesive display, translating complex ML outputs into immediately understandable financial insights for Gen-Z users.

This table presents the detailed use case description for the financial dashboard viewing functionality of **FinAnalyze**, which serves as the primary interface through which users visually interpret the results of their AI-generated financial analysis. The dashboard aggregates multiple interactive visualization components including spending category pie charts, monthly expenditure bar graphs, a behavioral radar chart, and a PCA scatter plot into a single, cohesive display that translates complex machine learning outputs into immediately understandable financial insights. The use case defines the system behavior for both data-available and data-unavailable states, ensuring that users receive clear and actionable guidance regardless of their current data submission status. This functionality is central to **FinAnalyze's** mission of making behavioral finance accessible and visually engaging for Pakistani Gen-Z users who may have no prior background in data science or personal finance.

Table 3.2.5 View Personalized Recommendations

Use Case ID	UC-5
Use Case Name	Interact with Robo-Advisor & FinBot
Actors	Primary: User; Secondary: Groq LLM API
Description	Provides highly personalized financial strategies and answers real-time economic queries using an integrated Large Language Model (LLM).
Trigger	User accesses the 'Robo-Advisor' tab.
Preconditions	PRE-1: A stable connection to the Groq API endpoint.
Postconditions	POST-1: Context-aware financial recommendations are displayed based on the user's specific K-Means personality cluster.
Business Rules	BR-1: Only anonymized, aggregated numerical ratios are transmitted to the Groq API. BR-2: The LLM must not receive any raw PII or bank account numbers.

This table outlines the administrative use case of **FinAnalyze**, which provides system administrators with a secure, role-restricted control panel for overseeing all aspects of platform operations including user account management, aggregated usage analytics, AI model version control, and comprehensive audit logging of all administrative actions. The use case defines the elevated authentication preconditions required for admin access, the four primary administrative functions available through the panel, and the exception handling procedures that ensure data integrity is preserved and rolled back safely in the event of server errors during any administrative operation. The analytics component of this use case enables administrators to monitor key platform metrics , such as total registered users, personality type distribution across the user base, and monthly active users , providing the data necessary to evaluate system performance and guide future development decisions. All administrative actions executed within this use case are permanently recorded in the audit log, ensuring full accountability, traceability, and compliance with the platform's data governance requirements.

Table 3.2.6 Admin Management and Analytics

Use Case ID	UC-6
Use Case Name	Admin Management and System Analytics
Actors	Primary: Admin
Description	Enables administrators to monitor system activity, oversee aggregate Firebase metrics, and handle live SMTP contact support tickets.
Trigger	Admin authenticates via the dedicated Admin Panel portal.
Preconditions	PRE-1: User must possess elevated 'Admin' role credentials.
Postconditions	POST-1: Platform health metrics and user distribution statistics are successfully rendered.
Business Rules	BR-1: Strict Role-Based Access Control (RBAC) must isolate the admin dashboard from standard users.

This outlines the secure, role-restricted control panel for overseeing platform operations. It enables administrators to monitor key platform metrics (total users, personality distribution) to evaluate system performance. It also ties into the Contact Support module, where user queries are routed via live SMTP integration to the administrative team.

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3.3 Functional Requirements

Additionally, the functional requirements also ensure that the system remains consistent, reliable, and user-centric throughout its operation. They act as a guideline for developers to implement each feature correctly while maintaining alignment with the overall project objective

Table 3.3 Functional Requirements Summary

FR-ID	Title	Requirement	Priority
FR-ID	Title	Requirement	Priority
FR-1	User Auth & Persistence	System shall securely log in users utilizing Bcrypt hashing and maintain session states (st.session_state) for seamless SPA navigation.	High

FR-2	Ephemeral Data Ingestion	System shall parse CSV/PDF uploads strictly in-memory (RAM), enforcing a Zero-Storage policy where raw credentials are permanently purged.	High
FR-3	ML Behavioral Analysis	System shall process numerical transaction ratios using Scikit-Learn (K-Means) to extract financial behavioral patterns.	High
FR-4	Cloud Data Storage	System shall store only the final categorized personality types and anonymized metrics in Firebase Firestore (NoSQL).	High
FR-5	Predictive Dashboard	System shall display interactive visualizations alongside a 10-year	Medium
		compounding Future Projection and Stress Test Simulator.	
FR-6	Context-Aware Robo-Advisor	System shall utilize the Groq LLM API to provide highly personalized financial planning tips based on the user's clustered personality.	High
FR-7	Administrative Monitoring	System shall provide an RBAC-protected Admin Panel to view aggregate platform analytics and user distributions.	High
FR-8	Live SMTP Support Routing	System shall process user feedback and queries via a native smtplib contact form, securely routing messages to admin inboxes.	Medium

3.4 Non-Functional Requirements

Non-Functional Requirements (NFRs) refer to the qualities and constraints that define how a system operates rather than what it does. They focus on aspects such as performance, usability, security, reliability, and scalability, ensuring that the system functions efficiently and provides a good user experience. While functional requirements describe the specific features of a system, non-functional requirements ensure that these features work smoothly, securely, and effectively under different conditions. They play a crucial role in making a system reliable, user-friendly, and suitable for real-world use by setting standards for speed, safety, and overall system quality.

3.4.1 Usability Requirements

- **Intuitive Navigation:** The interface shall be navigable without prior data science knowledge. A Single Page Application (SPA) design shall ensure fluid, app-like user experiences without full page reloads.
- **State Management:** The implementation of a Global Router (Priority-0) shall guarantee instant route resolution and seamless navigation with authenticated access control throughout the session.
- **Visual Hierarchy:** The system shall maintain a consistent layout utilizing the "Galixo" dark theme and glassmorphism CSS cards to reduce user cognitive load.
- **Interactive Feedback:** The system shall feature interactive tooltips, contextual error messages, and animated transitions to create a premium first impression during onboarding.

3.4.2 Performance Requirements

- **In-Memory Processing Speed:** The ML Analysis Hub shall execute data extraction via ephemeral PDF/CSV parsing without persistent disk I/O bottlenecks.
- **Real-Time Data Retrieval:** The integration of the Firebase Firestore NoSQL backend shall provide real-time, scalable data persistence, ensuring rapid loading of historical audit trails and user profiles.
- **UI Responsiveness:** Dashboard visualizations and interactive Plotly charts shall load instantly. Predictive modules shall switch between the 10-Year Wealth Trajectory and Stress Test Simulator without requiring page reloads.
- **API Latency Optimization:** External queries routed to the Groq LLM API for the Dual-AI assistants shall be optimized for minimal latency to maintain conversational fluidity.

3.4.3 Security Requirements

- **Zero-Storage Architecture:** All financial statement parsing shall be strictly memory-only,

leaving no persistent data trace. No sensitive raw data or banking credentials shall persist on disk or cloud storage ever.

- **Data Anonymization:** Before transmitting context to the external Groq LLM API, the system shall strip all Personally Identifiable Information (PII) to utilize anonymized AI prompts.
- **Cryptographic Protection:** The system shall utilize Bcrypt password hashing to provide enterprise-grade security for Login/Signup credentials.
- **Encrypted Communications:** All user feedback and support inquiries shall be routed through an SMTP-encrypted contact support channel to maintain confidentiality.
- **Regulatory Compliance:** The platform shall integrate Corporate Legal boilerplate.

3.4.4 Reliability Requirements

- **High Availability:** The system shall rely on Firebase Firestore's cloud infrastructure to maintain a minimum uptime of 99% during active academic deployment.
- **Auditability:** The system shall maintain complete, timestamped audit trails for all analytical activities via the Firebase History Log.
- **Fault Tolerance:** The Dual-agent architecture (Robo-Advisor and Finbot AI) shall ensure that users always have access to either specialized or general financial guidance depending on query complexity.
- **Safe Session Termination:** The secure logout mechanism shall guarantee the immediate termination of all active data and credentials from the browser state.

3.4.5 Maintainability Requirements

- **Modular Architecture:** Code shall be decoupled into distinct, independently testable modules mirroring the system architecture (Onboarding, Analysis Hub, Predictive Analytics, Dual-AI Assistants, User Management, and Administration).
- **Scalable Schema:** The utilization of a NoSQL database (Firestore) shall allow the platform to adapt to new user metrics or Persona structures without requiring rigid database migrations.
- **Monitoring Capabilities:** The exclusive Admin Panel shall provide role-based access control to monitor platform health metrics, API latency, error rates, and real-time user engagement.

3.4.6 Portability Requirements

- **Environment Agnosticism:** The system shall be seamlessly deployable across standard

cloud hosting platforms without modification to the core Streamlit logic.

- **Cross-Platform Compatibility:** The responsive web application shall function consistently across Windows, macOS, and Linux, ensuring accessibility across desktop and mobile browsers.

Chapter 4

Design and Architecture

4 Design and Architecture

This chapter describes the complete structural and behavioral design of FinAnalyze, including the system architecture, technology stack justification, process flows, ML algorithm design, UI/UX design principles, and all UML models.

The design follows the Model-View-Controller (MVC) architectural pattern, ensuring a clean separation of concerns between data management, application logic, and user presentation.

4.1 System Architecture

FinAnalyze is built on a three-layer client-server architecture following the Model-View-Controller (MVC) design pattern. The Presentation Layer, Application Logic Layer, and Data Persistence Layer each have distinct responsibilities with clean interfaces between them.

The Presentation Layer is implemented entirely in Streamlit with custom Glassmorphism HTML/CSS injected via `st.markdown()`. It handles all user interactions, form inputs, file uploads, chart interactions, toggle switches, and navigation. The sidebar navigation is clean and role-aware: the Admin tab appears only for admin users. The Forecasting page uses a horizontal toggle to present Standard Projection and Stress-Test Simulator within a single unified view, eliminating the need for a separate sidebar entry and giving the interface a professional SaaS quality.

The Application Logic Layer is pure Python. It contains the ML Engine (`k_means_engine.py`), the Preprocessing Module (`preprocessor.py`), the Business Rule Guardrails (`guardrails.py`), the Recommendation Engine (`recommendation_engine.py`), the Forecasting Engine (`forecasting.py`), the Authentication Module (`auth.py`), the SMTP Email Module (`email_handler.py`), and the FinBot module (`finbot.py`).

The Data Persistence Layer uses Firebase (`users.db`). Five tables are maintained: Users, Transactions, PersonalityProfiles, Recommendations, and AdminLogs. ML model files are serialized as pickle files stored on the server filesystem. SMTP credentials are stored securely in `.streamlit/secrets.toml` and accessed via `st.secrets`, never hardcoded in source files.

4.1.1 Presentation Layer (View)

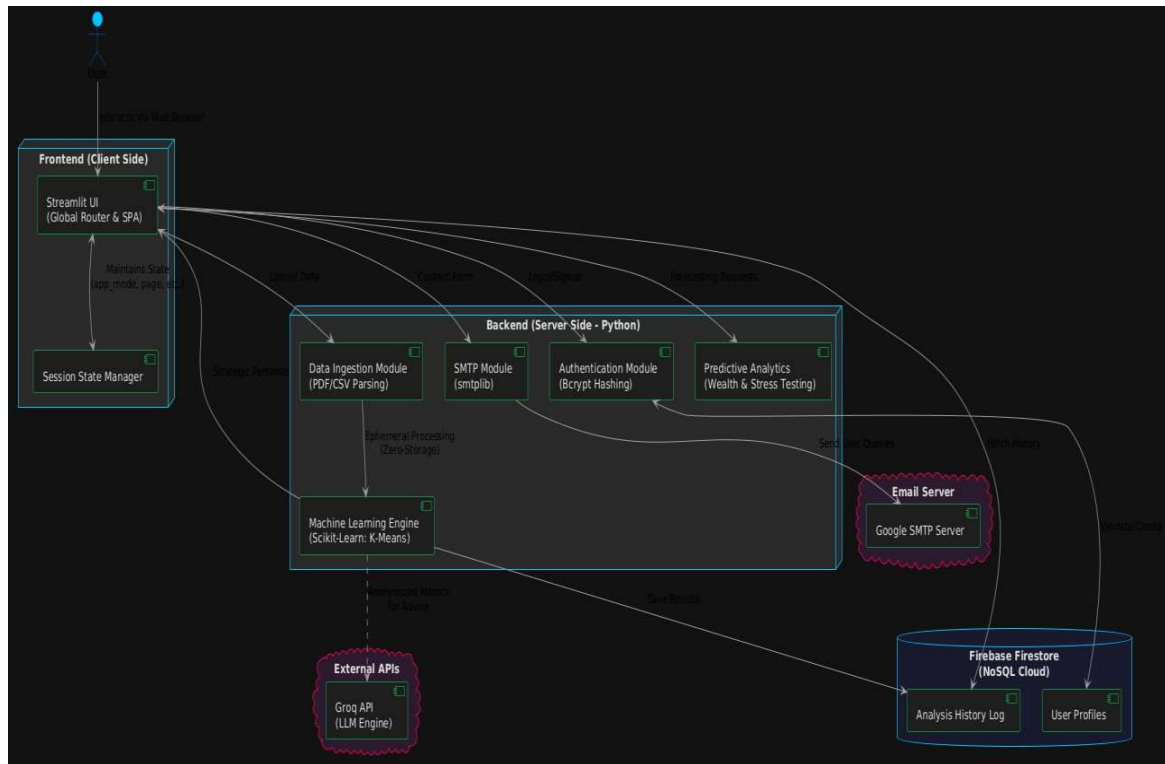
The Presentation Layer is responsible for all user-facing interactions. Built using Flask's Jinja2 templating engine combined with HTML5, CSS3, Bootstrap 5, and Plotly.js, this layer renders the following screens: Landing Page, Registration and Login forms, Financial Data Upload interface, Interactive Dashboard, Recommendations Panel, FinBot Chat Interface, User History Log, and Admin Control Panel. The UI is designed to be responsive across all screen sizes, with a dark/light mode toggle for accessibility.

4.1.2 Application Logic Layer (Controller/Model)

The Application Logic Layer is powered by Python 3.10+ with Flask handling HTTP routing and session management. It contains the Controller logic that processes user requests and routes them to the appropriate service. The ML Engine, implemented using Scikit-learn, performs the computationally intensive work of data preprocessing, K-Means and DBSCAN clustering, PCA dimensionality reduction, and personality label assignment. The Recommendation Engine uses a rule-based If-Else system to map personality types to curated financial tip sets. FinBot is powered by a Flask API endpoint that calls a language model for conversational responses.

4.1.3 Data Persistence Layer (Database)

Firebase serves as the relational database for **FinAnalyze**. The schema is designed around five core tables: Users (authentication and profile data), Transactions (uploaded financial records), Profiles (ML analysis results and personality labels), Recommendations (stored advice per session), and History (past analysis records for comparison). All tables are normalized to Third Normal Form (3NF) with indexed foreign keys for efficient query performance.



4.2 Technology Stack

The technology stack used in this project is carefully selected to ensure efficient development, robust scalability, and stringent data privacy. It combines modern programming tools and cloud infrastructure to support advanced machine learning, generative AI, and seamless web application development. Python is utilized as the core language for backend logic and ephemeral in-memory data processing, while libraries such as scikit-learn, pandas, and NumPy drive the core data manipulation and K-Means clustering algorithms for financial profiling [12], [14]. For the frontend and user interface, Streamlit is exclusively utilized [15] to construct a dynamic Single Page Application (SPA) with custom session-state routing, delivering a highly interactive web experience. Furthermore, the Groq API is integrated to power the context-aware conversational AI modules. Firebase Firestore is employed as the NoSQL cloud database for the secure, scalable retention of user profiles and historical logs, strictly adhering to the project's zero-storage policy for raw banking documents. This synergy of technologies ensures that the system remains highly secure, robust, and easy to maintain.

Table 4.2 Technology Stack Justification

Component	Technology	Justification
Component	Technology	Justification
Component	Technology	Justification
Web Framework & UI	Streamlit	Python-native UI framework ideal for data applications. Utilized to build a dynamic Single Page Application (SPA) with a custom Global Router and state management, bypassing complex frontend web stacks.
Backend Language	Python 3.10+	Core server-side language chosen for its robust ML library ecosystem, seamless API integrations, and efficient handling of in-memory data parsing logic.
Machine Learning Engine	Scikit-learn	Provides highly optimized implementations of machine learning algorithms, specifically utilized for K-Means Clustering to segment users into distinct "Financial DNA" personas.
Generative AI / LLMs	Groq API	Delivers ultra-low latency, state-of-the-art conversational AI capabilities. Powers the

Component	Technology	Justification
		context-aware 'Robo-Advisor' and 'Finbot AI' modules.
Data Processing (In-Memory)	Pandas, NumPy	Industry-standard libraries for rapid tabular data manipulation. Essential for the "Zero-Storage" architecture, allowing ephemeral PDF/CSV parsing in RAM without disk storage.
Cloud Database	Firebase Firestore (NoSQL)	Scalable, document-based NoSQL cloud database. Used strictly to persist user profiles, encrypted credentials, and historical analysis logs without storing raw banking documents.
Cryptography & Security	Bcrypt	Industry-standard cryptographic hashing algorithm implemented to securely salt and hash user passwords during the authentication and signup process.

4.3 Process Flow / Representation

The process flow of **FinAnalyze** describes the sequential data transformation journey from raw user input to final personalized output. Understanding this flow is essential for grasping how the system's components interact to produce meaningful results.

The process begins when a registered user uploads a CSV AND PDF file or enters transactions manually. The uploaded file is immediately validated for format correctness and column

completeness. Upon successful validation, the Backend Parser reads the file using pandas, converting the raw spreadsheet data into a structured DataFrame with standardized column names for Date, Amount, Category, and Description.

The Preprocessing Module then takes this DataFrame through a series of data cleaning and transformation steps. Null values in numeric columns are filled with column means, while rows with missing categorical values are dropped. Date strings are converted to datetime objects. The Category column is one-hot encoded to create binary indicator columns for each spending category: Food, Bills, Entertainment, Shopping, Investment, and Other. The system then computes aggregate features per user: total spending per category, spending frequency, average transaction amount, and Savings Ratio calculated as (Income minus Total Expenses) divided by Income. All numerical features are then normalized using StandardScaler, centering each feature at a mean of zero with a standard deviation of one.

The normalized feature vector is passed to the ML Engine. K-Means clustering assigns the user's data point to one of k clusters, where k is determined using the Elbow Method applied during model training. PCA is then applied to the same feature vector, reducing it to two principal components for dashboard visualization. The cluster assignment is mapped to a human-readable personality label using a predefined mapping table.

The personality label is stored in the database and passed to the Recommendation Engine, which retrieves the corresponding set of financial recommendations. The Dashboard Module assembles all visualization components, pie chart, bar graph, radar chart, PCA scatter plot, and personality badge, and renders them in the user interface. The complete analysis cycle, from data submission to results display, is designed to complete within 30 seconds for standard dataset sizes.

After data processing, the application's flow divides into three parallel strategic modules. The first module is the Strategic Dashboard, where interactive Plotly charts render the user's financial health metrics. The second module is Predictive Analytics, which calculates a 10-year wealth trajectory based on the data and runs a Stress Test simulator for scenarios like market crashes or inflation. The third module consists of the Dual-AI Assistants; if the query is related to personalized data, the Groq API-powered Robo-Advisor responds, whereas general financial questions are handled by the Finbot AI. Finally, when the user ends the session, a Session-Safe Logout is activated, which completely purges any remaining active memory and credentials in the background, ensuring total privacy is maintained.

FinAnalyze Pro - Core Process Flow

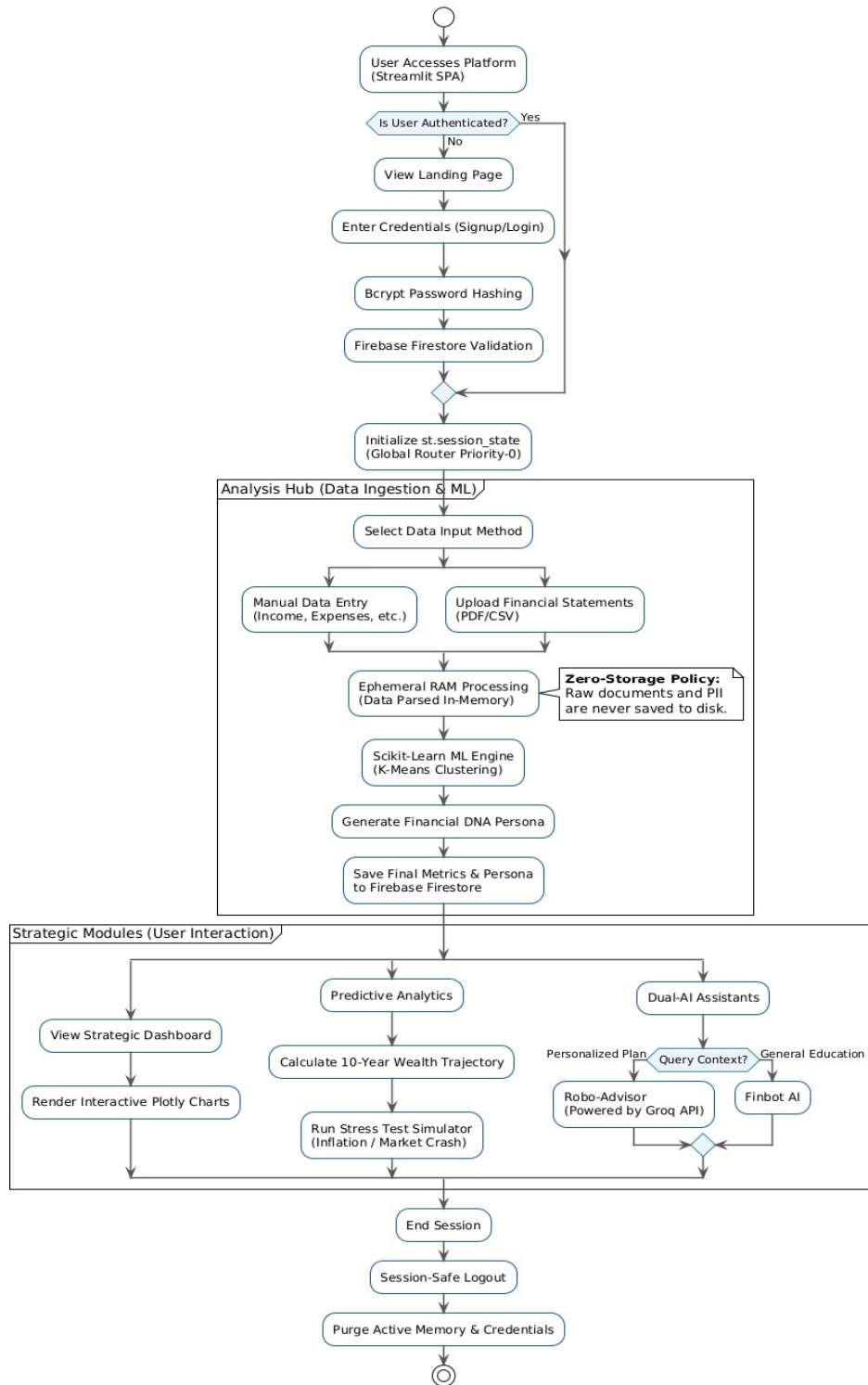


Figure 4.2 FinAnalyze Process Flow Diagram

4.4 Sequence Diagram

Phase 1: Secure Onboarding

The sequence begins with the user entering their credentials (login or signup) into the Streamlit Single Page Application (SPA). The SPA sends this data to the Auth Module, which hashes and validates the password using Bcrypt. This ensures enterprise-grade security for credential protection. The Auth Module then verifies the user's document within the Firebase Firestore database. Upon receiving a success response from the database, the Auth Module initializes the `st.session_state` to maintain the user's session securely. Finally, the SPA renders the Dashboard or Landing page for the user.

Phase 2: Zero-Storage Analysis (Data Ingestion)

In this core phase, the user uploads a financial statement (PDF or CSV) to the Streamlit SPA. The application sends the raw file bytes directly to the In-Memory Parser. A critical note on the diagram highlights the system's "Zero-Storage Policy," explicitly stating that data resides purely in RAM and no files are ever saved to disk. This aligns with the platform's ephemeral data extraction and memory-only processing approach.

The parser extracts the necessary variables and vectors, then sends these abstracted vectors to the Machine Learning (ML) Engine. The Scikit-Learn ML Engine executes K-Means Clustering to identify the user's "Financial DNA." Crucially, the ML Engine saves *only* the final calculated persona and numerical metrics to Firebase Firestore. Once the database acknowledges the save, the ML Engine returns the strategic persona data to the SPA. At this exact moment, the raw statement data is immediately purged from the RAM to ensure total privacy. The SPA then displays an interactive health dashboard to the user.

Phase 3: Conversational AI (Robo-Advisor)

This phase outlines the data flow for the personalized Robo-Advisor. The user asks a financial strategy question through the SPA interface. To provide context-aware advice, the SPA fetches the user's saved persona metrics from Firebase Firestore. The SPA then combines the user's question with this context and sends it to the Groq API (LLM). A security note emphasizes that all Personally Identifiable Information (PII) is stripped before querying the external AI service; only numerical and persona context is transmitted. The Groq API processes the anonymized prompt and returns tailored AI advice, which the SPA then displays as a personalized financial strategy to the user.

Phase 4: Session Termination

The final phase covers the secure exit process. The user initiates this by clicking "Logout" on the Streamlit SPA. In response, the SPA immediately purges the `st.session_state` and clears the active memory. This session-safe logout guarantees that all active data and banking credentials are comprehensively terminated. Following the memory purge, the user is safely redirected back to the initial login screen.

Overall, this sequence diagram serves as a powerful visual proof of the system's robust architecture, perfectly balancing advanced AI capabilities with uncompromising data privacy. By clearly mapping out the journey from the initial encrypted login to the ephemeral in-memory processing and the final session-safe logout, it demonstrates to the defense panel that FinAnalyze Pro is not just a functional prototype, but a thoughtfully engineered, enterprise-grade platform. The strict adherence to the Zero-Storage policy and the anonymization of LLM prompts highlight a deep understanding of modern cybersecurity and data compliance, ensuring that your project addresses real-world financial tech challenges with utmost professionalism.

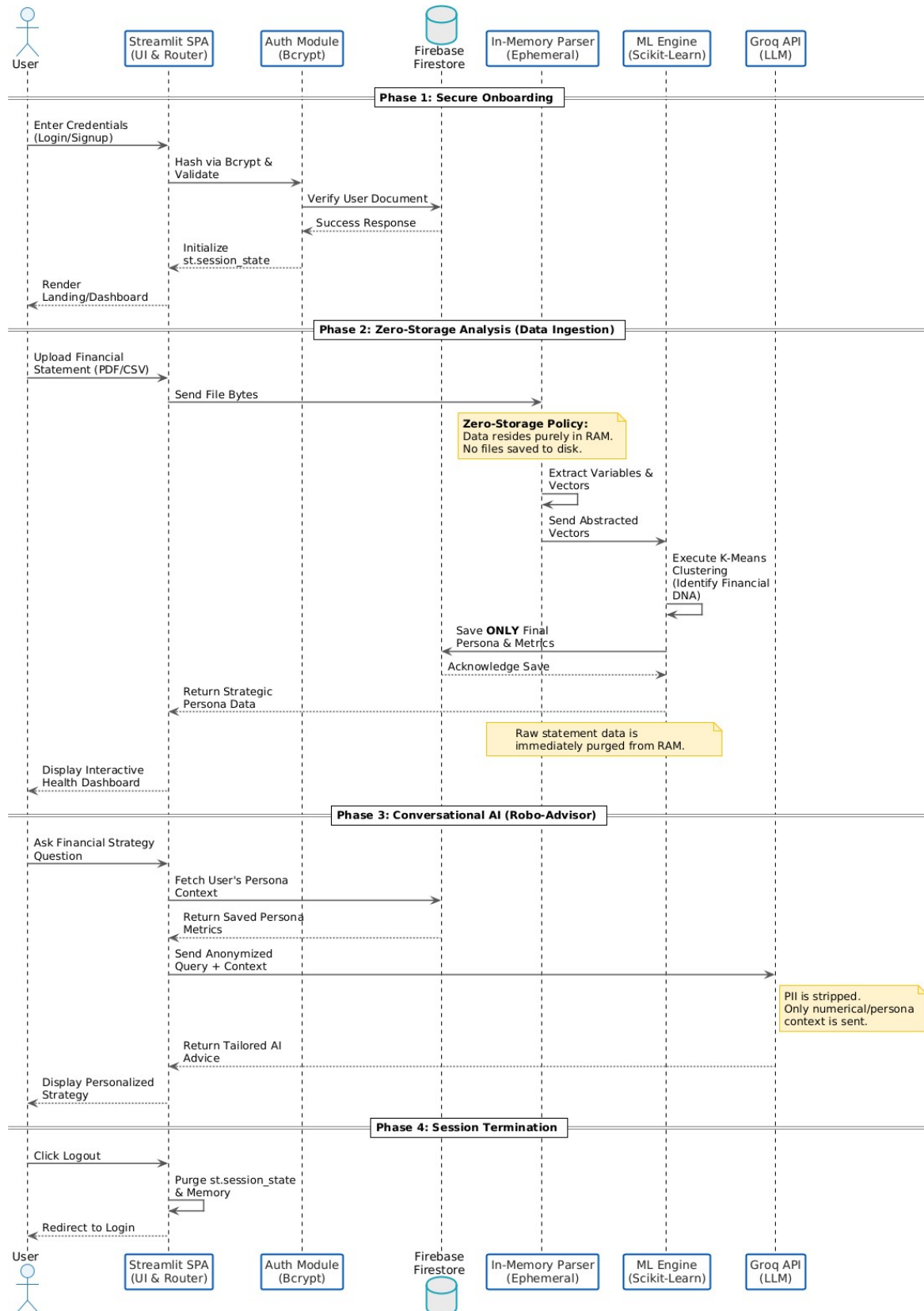


Figure 4.3 FinAnalyze Sequence Diagram

4.5 Class Diagram

The class diagram for **FinAnalyze** represents the static structural organization of the system's primary software components using an object-oriented model. It defines the key classes, their attributes, methods, and the relationships between them.

The User class is the central entity, containing attributes for `user_id`, `full_name`, `email`, `password_hash`, `registration_date`, and `role`. Its methods include `register()`, `login()`, `logout()`, `update_profile()`, and `get_history()`. The User class has a one-to-many relationship with the Transaction class, as each user may have multiple transaction records.

The Transaction class contains attributes for `transaction_id`, `user_id` (foreign key), `date`, `amount`, `category`, and `description`. Its primary method is `get_transactions_by_user()`, which retrieves all transactions associated with a given `user_id`.

The MLEngine class encapsulates the machine learning processing logic. Its attributes include the trained model (`kmeans_model`), the PCA transformer (`pca_model`), and the feature list

The PersonalityProfile class stores the results of each ML analysis. Attributes include `profile_id`, `user_id` (foreign key), `personality_type`, `pca_component_1`, `pca_component_2`, `savings_ratio`, `top_category`, and `analysis_date`. The class has a many-to-one relationship with the User class.

The RecommendationEngine class is responsible for generating personalized advice. Its `get_recommendations(personality_type)` method applies rule-based logic to return a list of Recommendation objects corresponding to the given personality archetype.

The Admin class extends the User class with additional methods: `manage_users()`, `view_analytics()`, and `update_model_version()`. The FinBot class handles chatbot interactions, with a `respond_to_query(user_input)` method that processes natural language questions and returns contextual financial guidance.

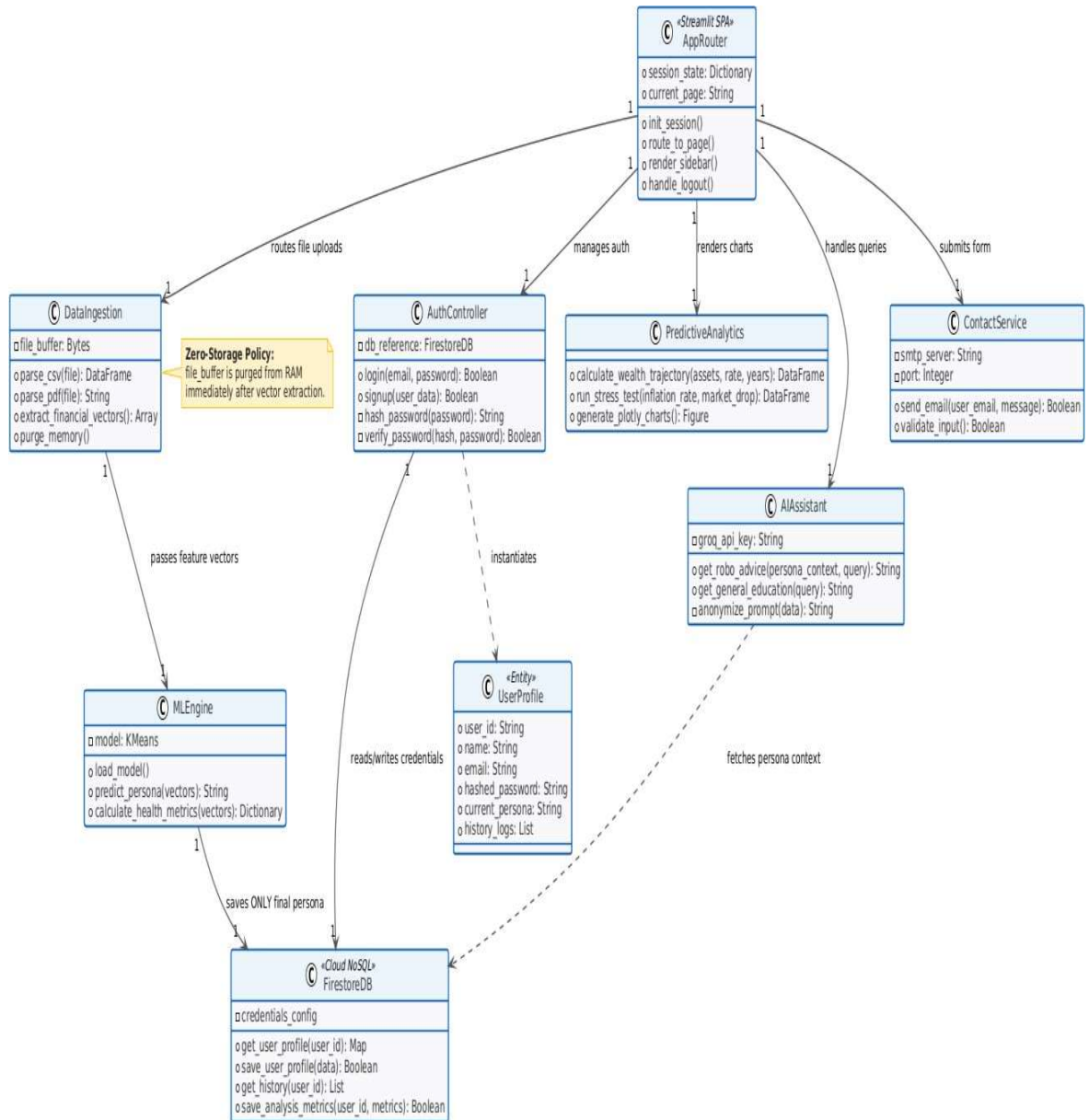


Figure 4.4 FinAnalyze Class Diagram

4.6 Database Schema

Table 4.6 presents the overall database schema of **FinAnalyze**, integrating multiple interconnected tables to support user-centric financial analysis. The Users table serves as the core entity, storing authentication and identity details, while the Transactions table captures detailed financial activities linked to each user for behavioral analysis. The Profiles table generates and maintains user-specific financial behavior profiles, including personality labels and analytical metrics, which

are further utilized by the Recommendations table to provide personalized financial advice. Additionally, the History table preserves past user profiles and snapshots, enabling longitudinal tracking of behavioral changes over time. Together, these tables form a structured and relational data model that supports data storage, analysis, personalization, and continuous monitoring within the FinAnalyze system.

Table 4.6 Database Schema Summary

Table	Primary Key	Foreign Keys	Key Attributes
Users	user_id (INT)	,	email, password_hash, role, created_at
Transactions	transaction_id (INT)	user_id → Users	date, amount, category, description
Profiles	profile_id (INT)	user_id → Users	personality_label, cluster_id, savings_ratio, generated_at
Recommendations	rec_id (INT)	profile_id → Profiles	tip_text, tip_category, display_order
History	history_id (INT)	user_id → Users	profile_id, snapshot_date, personality_label

4.7 Data Design

The data design section defines the logical structure of the system's relational database, which is implemented using Firebase. The database is carefully designed to ensure data consistency, scalability, and efficient retrieval of user financial information. It follows normalization principles (up to 3NF where applicable) to eliminate redundancy and maintain referential integrity across all tables.

The **Users table** is the master table and stores all registered user information. It includes the following fields: user_id (INTEGER, PRIMARY KEY, AUTOINCREMENT), full_name (TEXT, NOT NULL), email (TEXT, UNIQUE, NOT NULL), password_hash (TEXT, NOT NULL), role

(TEXT, DEFAULT 'user'), and registration_date (DATETIME, DEFAULT CURRENT_TIMESTAMP). Passwords are stored in hashed format to ensure security and prevent unauthorized access.

The **Transactions table** records all financial activities performed or uploaded by the user. It includes: transaction_id (INTEGER, PRIMARY KEY, AUTOINCREMENT), user_id (INTEGER, FOREIGN KEY referencing Users.user_id), transaction_date (DATE, NOT NULL), amount (REAL, NOT NULL, CHECK amount > 0), category (TEXT, NOT NULL), and description (TEXT). This table acts as the core dataset for machine learning analysis.

The **PersonalityProfiles table** stores outputs generated by the clustering and PCA algorithms. It includes: profile_id (INTEGER, PRIMARY KEY, AUTOINCREMENT), user_id (INTEGER, FOREIGN KEY), personality_type (TEXT, NOT NULL), savings_ratio (REAL), top_category (TEXT), pca_x (REAL), pca_y (REAL), and analysis_date (DATETIME, DEFAULT CURRENT_TIMESTAMP). This table enables historical tracking of user financial behavior over time.

The **Recommendations table** contains predefined financial advice mapped to personality types. It includes: recommendation_id (INTEGER, PRIMARY KEY), personality_type (TEXT, NOT NULL), recommendation_text (TEXT, NOT NULL), and category_tag (TEXT). This structure allows fast retrieval of relevant suggestions without requiring real-time computation.

The **AdminLogs table** provides an audit trail for system monitoring and security. It includes: log_id (INTEGER, PRIMARY KEY, AUTOINCREMENT), admin_id (INTEGER, FOREIGN KEY), action_type (TEXT), target_entity (TEXT), and timestamp (DATETIME, DEFAULT CURRENT_TIMESTAMP). This ensures accountability for administrative actions and supports system debugging and analytics.

Additionally, indexing is applied on frequently queried fields such as user_id and transaction_date to improve query performance. Foreign key constraints are enforced to maintain relational integrity between tables. The database design also supports future scalability, allowing new tables such as notifications, chatbot logs, or advanced analytics storage to be integrated without restructuring the existing schema.

4.8 Algorithm Design

The algorithmic core of the system consists of three interconnected stages: data preprocessing and cleaning, AI-based clustering using K-Means and DBSCAN, and a rule-based recommendation engine. These stages collectively transform raw financial data into meaningful insights and personalized guidance.

The first stage involves **data preprocessing**, where raw transaction data is cleaned, normalized, and transformed into a machine-readable format. This ensures that the dataset is free from inconsistencies and suitable for machine learning algorithms.

The second stage involves **unsupervised machine learning**, where clustering techniques such as K-Means and DBSCAN are applied to identify hidden patterns in user financial behavior. These algorithms group users based on similarities in spending habits, income distribution, and savings behavior.

The third stage involves a **rule-based recommendation system**, which maps each identified personality type to predefined financial advice. This ensures that insights generated by the machine learning model are translated into practical and actionable recommendations for users.

Together, these three stages form a complete pipeline that enables the system to move from raw data input to intelligent financial decision support, ensuring both analytical depth and user usability.

4.8.1 Data Preprocessing and Cleaning

The preprocessing algorithm begins by reading the uploaded CSV AND PDF AND file into a pandas DataFrame. The algorithm checks for null values in the Amount column and fills them with the column mean. Rows with missing Category values are dropped entirely to preserve categorical data integrity. The Date column is converted from string format to pandas Timestamp objects using the `to_datetime()` function. Data is then grouped by Category to compute the total spending per category for each user. The Savings Ratio is calculated as (Income minus Total Expenses) divided by Income. One-hot encoding is applied to the Category column, creating a binary indicator column for each spending category. Finally, StandardScaler is applied to all numerical features, normalizing them to zero mean and unit standard deviation, ensuring that features with larger absolute values do not disproportionately influence the clustering distance calculations.

Additionally, outlier detection is performed using the Interquartile Range (IQR) method to remove extreme transaction values that may distort clustering results. Duplicate records are also identified and removed to maintain dataset consistency. Feature selection is applied to retain only relevant financial attributes such as income, expenses, category distribution, and savings ratio. This ensures that the model is trained on meaningful and non-redundant data, improving overall clustering accuracy.

4.8.2 K-Means Clustering Algorithm

The K-Means algorithm is used as the primary unsupervised learning technique to segment users based on their financial behavior patterns [6], [7]. It operates by partitioning the dataset into k distinct clusters, where each cluster represents a unique financial personality type. The optimal number of clusters (k) is determined using the Elbow Method, which evaluates the Within-Cluster Sum of Squares (WCSS) across different values of k and identifies the point where adding more clusters results in diminishing returns.

Each user is represented as a multidimensional feature vector consisting of attributes such as total income, total expenses, savings ratio, category-wise spending distribution, and transaction frequency. The algorithm initially selects random centroids and assigns each user to the nearest centroid based on Euclidean distance. After assignment, centroids are recalculated as the mean of all points within each cluster.

This iterative process continues until convergence is achieved, meaning that centroid positions no longer change significantly or a maximum number of iterations is reached. This ensures stability and consistency in clustering results.

To improve interpretability and visualization, Principal Component Analysis (PCA) is applied after clustering. PCA reduces high-dimensional financial data into two principal components (PC1 and PC2), enabling graphical representation of clusters in a two-dimensional space. This helps users visually understand their position relative to other financial behavior groups.

Model evaluation is performed using the Silhouette Score, which measures intra-cluster cohesion and inter-cluster separation. A higher silhouette score indicates that users within a cluster are more similar to each other while being distinctly different from users in other clusters.

Additionally, multiple initialization runs (k-means++) are used to improve centroid selection and avoid local minima. This enhances clustering stability and ensures more reliable segmentation of financial personalities

4.8.3 Recommendation Engine Logic

The recommendation engine is implemented as a rule-based decision system that maps each identified financial personality type to a structured set of personalized financial advice. It acts as the interpretability layer of the system, converting machine learning outputs into actionable insights for users.

The engine operates using conditional logic (if-else rules) based on the predicted cluster label. Each personality type is associated with a predefined knowledge base of financial recommendations tailored to user behavior patterns.

For example, users classified as *The Smart saver*, receive recommendations focused on investment opportunities, compound interest benefits, emergency fund planning, and long-term wealth accumulation strategies. Users identified as *The Big spender* are provided with behavioral control techniques such as budgeting frameworks (e.g., 50/30/20 rule), spending alerts, and impulse reduction strategies.

Similarly, *The Budget challenger* category is guided toward diversified investment portfolios, risk assessment techniques, and financial planning strategies to balance high-risk behavior. *The Wealth builder* users receive optimization-focused advice that helps them maintain financial efficiency while ensuring sustainable savings habits.

The system ensures that each user session generates a minimum of three recommendations, ranked based on relevance and behavioral severity. These recommendations are dynamically retrieved from a structured database to ensure fast response time and scalability.

Furthermore, recommendation logic is designed to be modular, allowing easy extension for future enhancements. In advanced versions, this rule-based system can be replaced or supplemented with machine learning-based recommendation engines, such as collaborative filtering or reinforcement learning models.

Future improvements also include context-aware recommendations, where suggestions adapt dynamically based on real-time spending behavior, seasonal patterns, and historical financial trends. Integration with natural language processing could further enable conversational recommendation delivery through the FinBot assistant, making financial guidance more interactive and personalized.

Chapter 5

Implementation

5 Implementation

This chapter describes the practical implementation of FinAnalyze Pro across all its constituent modules. Each section corresponds to a functional component of the system and discusses the development approach, key technical decisions, and implementation details necessary for the realization of this AI-powered financial profiling platform.

5.1 Development Environment Setup

The development environment for this project is built on modern, open-source tools compatible with robust data science and web application infrastructure. Python 3.10+ serves as the primary programming language, selected for its dominance in machine learning applications [12], [14]. Visual Studio Code was used as the primary integrated development environment, leveraging Python extensions for code intelligence and linting.

All Python package dependencies are managed through pip and declared in a requirements.txt file. The primary packages include pandas for tabular data manipulation, scikit-learn for machine learning algorithms [12], [14], numpy for numerical computation, and plotly for interactive data visualization. The user interface and application routing are constructed entirely using streamlit [15], operating as a Single Page Application (SPA). firebase-admin is utilized for secure, scalable cloud database operations.

Version control was rigorously maintained using Git with a remote repository hosted on GitHub. Each development sprint corresponded to a separate feature branch, which was merged into the main branch upon completion and rigorous testing.

5.2 Cloud Database Implementation (Firestore)

Diverging from traditional relational databases, FinAnalyze Pro implements a NoSQL document-oriented database using Firebase Firestore. This architecture provides scalable, lightweight, and efficient data storage tailored for modern web applications. The database is structured around discrete collections, primarily users and analysis_history.

Connection management is securely handled via the firebase-admin SDK using a service account key, ensuring authenticated server-to-server communication. The NoSQL structure allows for flexible schema evolution. When a user account is created, a unique document ID is generated. In adherence to the platform's stringent privacy standards, the database stores *only* user profiles,

hashed credentials, and the final computed metrics (such as the assigned Financial Persona and health scores). It strictly prohibits the persistence of raw transaction logs.

5.3 User Authentication and Authorization

The authentication system implements a secure onboarding flow that collects the user's full name, email address, and password. Critical to the system's security posture, passwords are mathematically hashed using the bcrypt algorithm with automated salting before storage, ensuring that plaintext passwords are never exposed or written to the database.

Session management and access control are implemented using Streamlit's native `st.session_state` mechanism, functioning as a Global Router (Priority-0) [15]. This maintains the user's authenticated state across the SPA without requiring full page reloads. The system distinguishes between standard user accounts and administrative accounts via a role attribute within the Firestore user document, ensuring the Admin Panel remains strictly isolated and accessible only to authorized personnel.

5.4 Data Ingestion and Zero-Storage Architecture

The Data Ingestion Module handles both file upload (PDF/CSV) and manual entry pathways. For file uploads, the module utilizes `pandas read_csv()` and specific PDF parsing libraries to extract financial data directly into an in-memory DataFrame [12].

A rigorous validation function ensures the presence of required columns (Date, Amount, Category), verifies that numerical values are valid, and standardizes datetime formats. Validation errors are returned to the user with actionable feedback. Manual entry is facilitated through intuitive Streamlit form inputs featuring client-side validation to prevent malformed data injection.

Crucially, this module implements the "Zero-Storage Policy." Upon successful parsing and validation, the standardized dataset is held temporarily in RAM (Random Access Memory). Once the data is passed to the machine learning pipeline for feature extraction, the raw file buffers and dataframes are immediately purged from memory. This ephemeral processing guarantees that raw banking documents and Personally Identifiable Information (PII) are never saved to disk, eliminating the risk of data breaches.

5.5 Machine Learning Engine Implementation

The ML Engine encapsulates the core data science logic, orchestrating the pipeline from feature extraction to personality classification. Feature engineering computes essential vectors for each user, including total category spending (food, bills, entertainment), spending frequency, average transaction size, and savings ratio. These features form the numerical input vector for the clustering algorithm.

The core profiling mechanism utilizes the K-Means Clustering algorithm [12], [14]. This model is pre-trained on aggregated, anonymized financial datasets to establish baseline behavioral segments. The Elbow Method analysis was utilized during training to identify the optimal number of clusters (\$k\$), corresponding to distinct financial personality archetypes (e.g., The Smart saver,, The Big spender).

At inference time, the user's ephemeral feature vector is processed through the Scikit-Learn pipeline, assigning them to the nearest cluster centroid to map their unique 'Financial DNA'. Only the resulting classification and aggregated numerical metrics are retained for database storage.

5.6 Strategic Visualization Dashboard

The Strategic Dashboard is the primary user-facing output, rendered dynamically via Streamlit [15]. It transforms complex data into intuitive visual components:

- **The Personality Card:** Prominently displays the user's identified financial archetype, complete with visual styling and a definitive tag.
- **Health Metrics:** Key Performance Indicators (KPIs) such as the Savings Ratio are displayed numerically with color-coded health indicators (green for optimal, amber for moderate, red for critical).
- **Interactive Visualizations:** Utilizing Plotly [13], the dashboard renders interactive charts, such as spending category breakdowns and monthly expenditure trends, allowing users to hover over data points for exact values.

This dashboard provides an immediate, comprehensive overview of the user's financial standing without compromising the underlying zero-storage architecture.

5.7 Predictive Analytics & Future Projections

Beyond immediate profiling, FinAnalyze Pro incorporates a Predictive Analytics module. This

component projects a 10-Year Wealth Trajectory based on the user's current savings rate and assumed compounding interest.

Furthermore, it integrates a Stress Test Simulator, allowing users to model the resilience of their projected wealth against adverse economic scenarios, such as high inflation rates or sudden market downturns. These predictive models are visualized using interactive line charts, providing users with a tangible understanding of their long-term financial trajectory and risk exposure.

5.8 Dual-AI Assistants (Robo-Advisor & Finbot)

FinAnalyze Pro implements a dual-agent conversational AI architecture, accessible directly from the dashboard, leveraging the high-speed Groq API (LLM engine).

1. **Robo-Advisor (Context-Aware):** This agent provides highly personalized strategic recommendations. Before querying the Groq API, the system fetches the user's saved numerical metrics and Persona type from Firestore. Crucially, an anonymization layer strips all PII from the prompt. The Groq LLM receives only the behavioral context, returning tailored advice on budgeting, investment, or savings specific to the user's Financial DNA.
2. **Finbot AI (General Education):** This secondary agent functions as a general financial educator. It handles natural language queries regarding broad financial concepts (e.g., "What is compound interest?" or "How do mutual funds work?"), providing instant, accessible financial literacy without requiring user-specific data context.

5.9 Administrative Control Panel

The Admin Panel is an exclusive route protected by role-based access control, designed for system oversight and governance. It provides administrators with a centralized view of platform health and engagement.

The panel aggregates platform statistics by executing read-only queries against the Firebase Firestore collections. It displays metrics such as total registered users, the distribution of personality type classifications across the platform, and aggregate system usage. It also provides secure tools for User Management, allowing administrators to monitor active accounts and manage system access, ensuring the platform remains secure and functional at scale.

5.10 User Interface Screens

The user interface of FinAnalyze Pro is designed with a modern, wealth builder 'Galixo' dark theme, optimizing for clarity, professional aesthetics, and the specific expectations of a digital-native demographic. The UI is constructed as a seamless Single Page Application.

The initial entry point is the Landing Page, featuring smooth animations, a clear value proposition ("AI-Powered Financial DNA Decoding"), and a secure portal for onboarding. Once authenticated, users navigate a persistent sidebar that routes them between the Data Ingestion Hub, the Strategic Dashboard, the Predictive Analytics center, the AI Assistant interface, and their secure Profile settings, ensuring a cohesive and intuitive user experience.

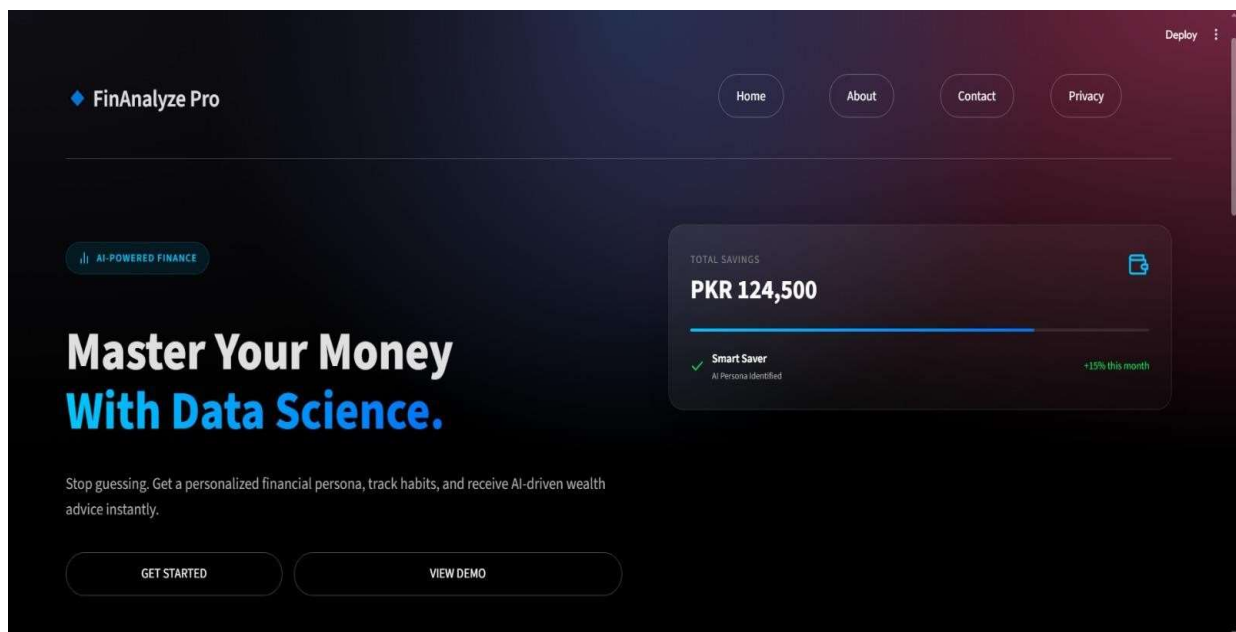


Figure 5.10.5 FinAnalyze Landing Page 1

The Registration and Login screens provide clean, minimal forms for user authentication. The registration form collects full name, email, and password, and initiates OTP verification upon submission. The login form allows returning users to access their dashboard with email and password credentials.

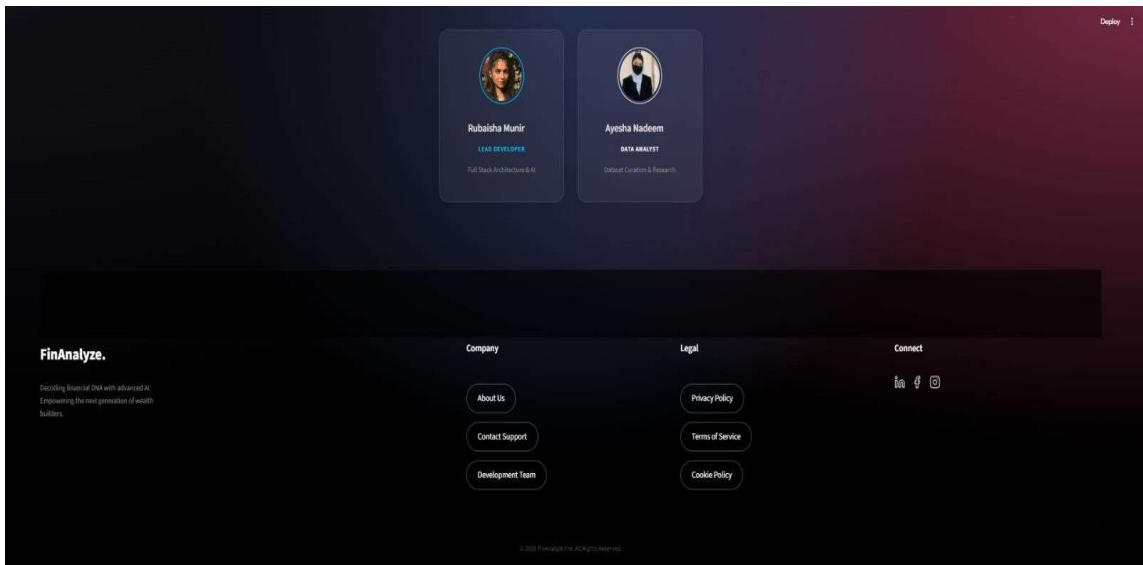


Figure 6.10.2 FinAnalyze Landing Page 2

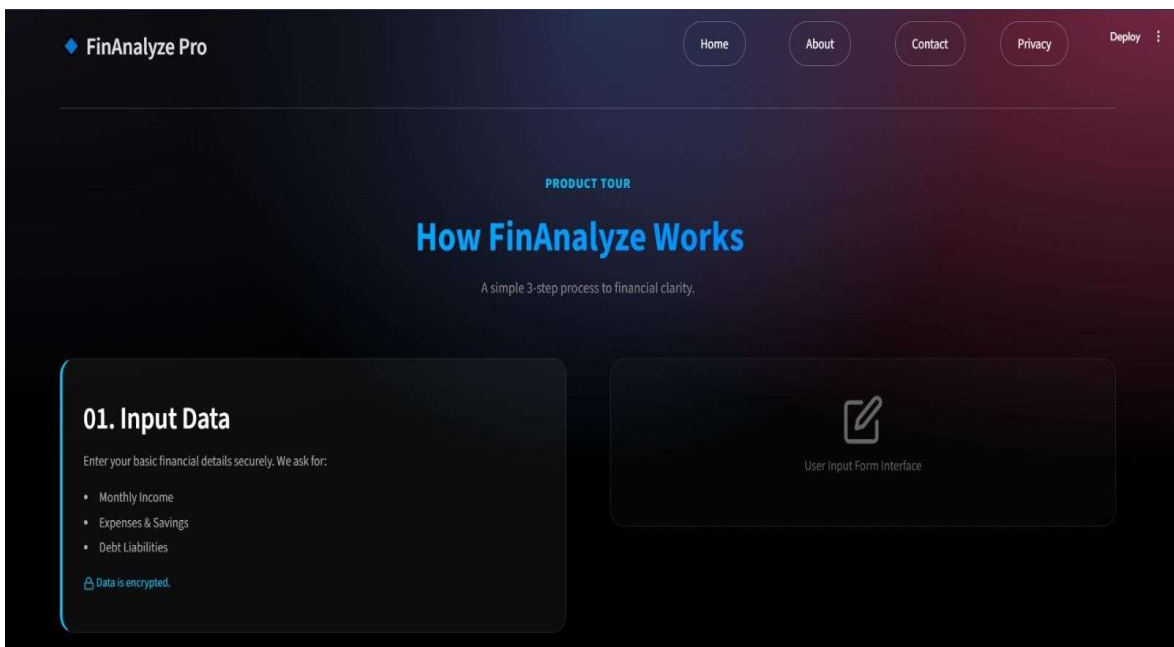


Figure 7.10.3 FinAnalyze View Demo

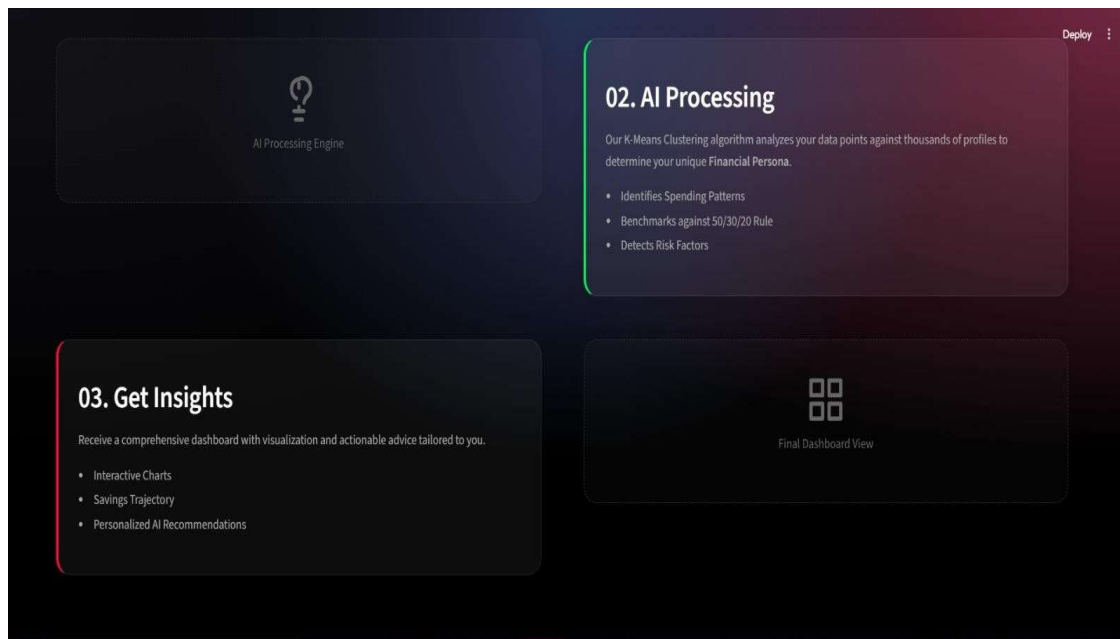


Figure 8 .10.4 Demo page 2

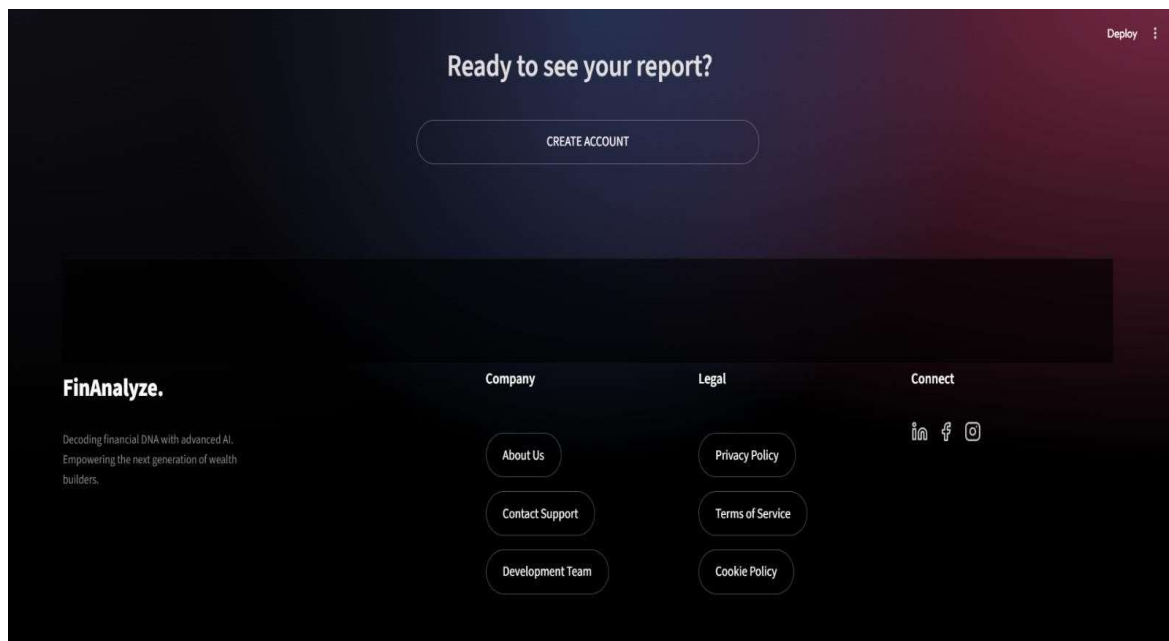


Figure 9 .10.5 Demo 3

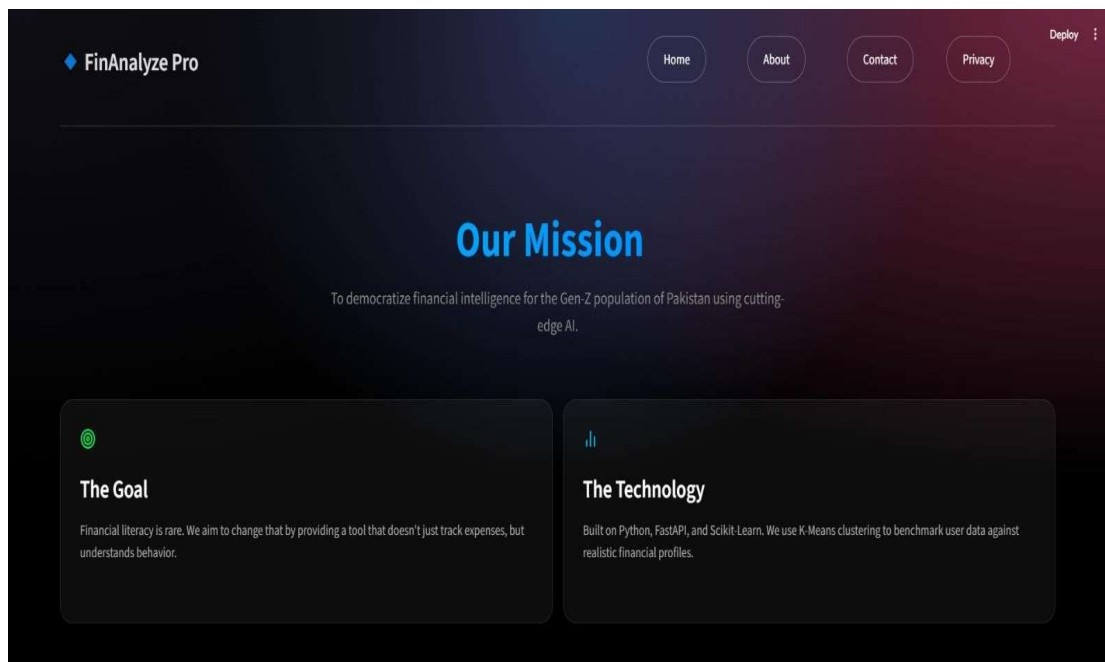


Figure 10.10.6 About us

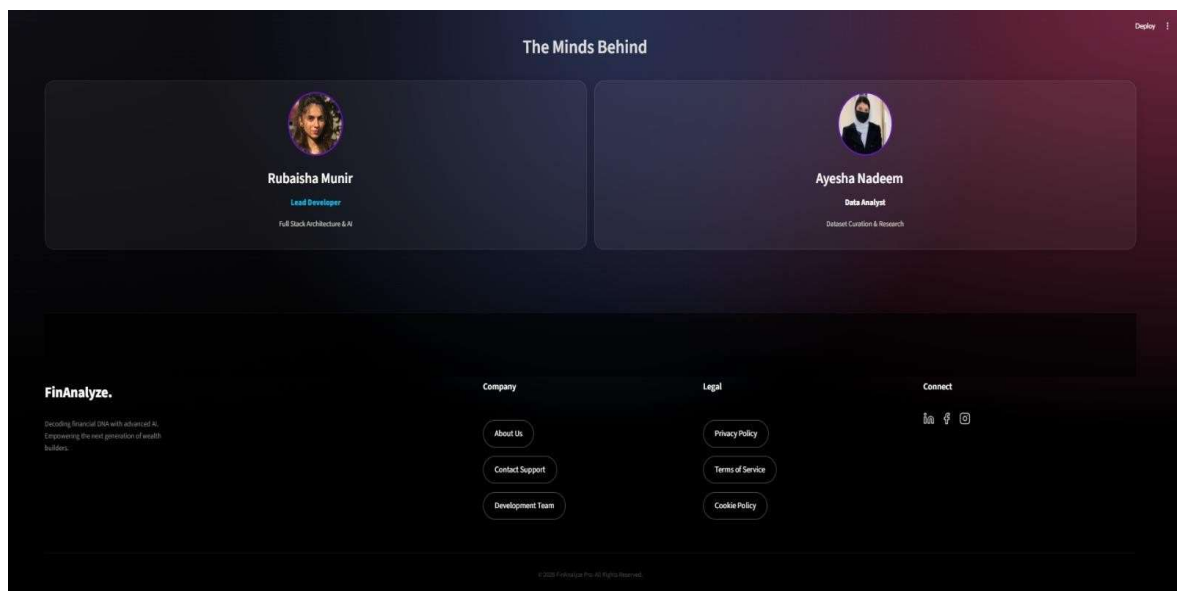


Figure 11.10.7 About us page 2

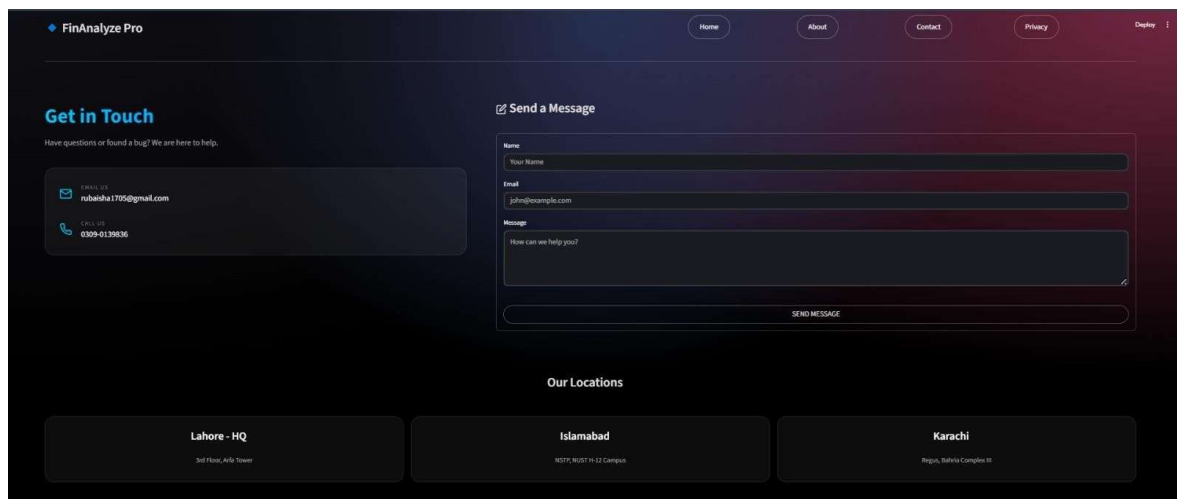


Figure 12 .10.8 Contact us

Figure 13 .10.9 Register

The Data Input Screen provides two pathways: a file upload zone accepting CSV AND PDF or XLSX files with drag-and-drop functionality, and a manual transaction entry form with fields for Date (calendar picker), Amount (numeric field), Category (dropdown: Food, Bills, Entertainment, Shopping, Investment, Other), and Description (optional text).

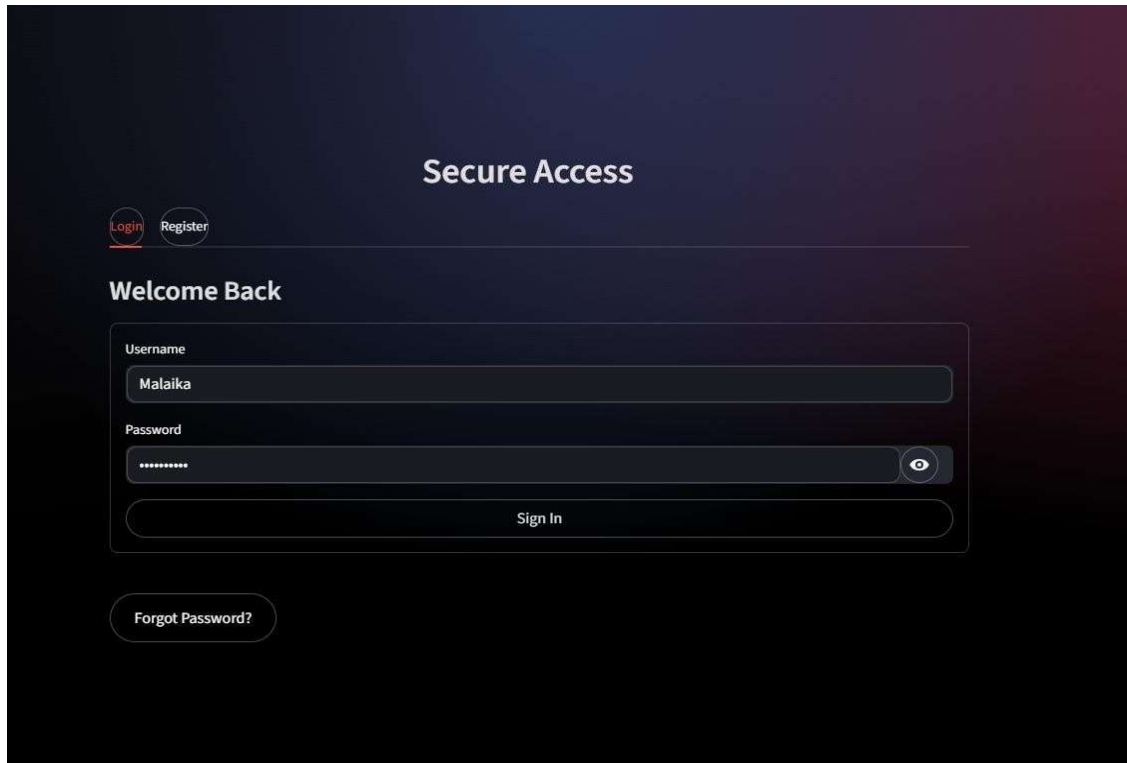


Figure 14 .10.10 Secure Access



Figure 15 .10.11 User dashboard

FinAnalyze Enterprise
Analysis Dashboard for Malaika Arshad

Data Entry Interface

Select Input Method:
☒ Manual Entry ☐ Upload Bank Statement (CSV / PDF)

Monthly Income (PKR): 100000
 Monthly Expenses (PKR): 50000
 Total Savings (PKR): 20000
 Outstanding Debt (PKR): 0

Spending Breakdown
 Essentials (Rent, Food, Bills): 30000
 Lifestyle (Shopping, Outings): 20000

Execute Analysis

Navigation:
 Analysis Hub
 Edit Profile
 FinBot AI
 Logout

Main Menu:
 Overview / Dashboard
 New Analysis
 Future Projections
 Robo-Advisor
 History Log

Back to Home

Figure 16.10.12 Manually data entry

The AI Analysis Results Screen is the core output display, presenting the user's financial personality badge prominently at the top, followed by the Savings Ratio KPI, the radar chart, the spending category pie chart, and the PCA scatter plot showing the user's position relative to all four personality cluster centroids.

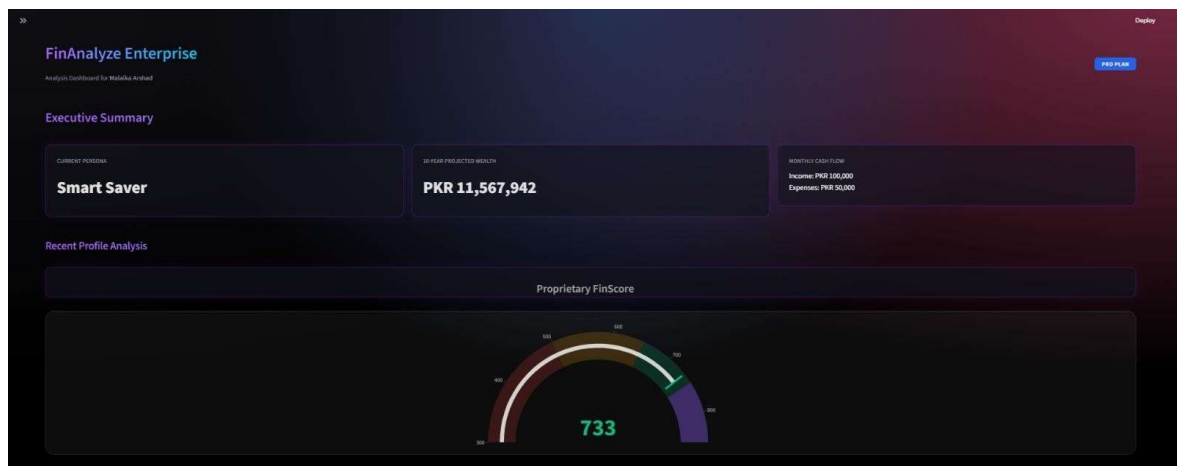


Figure 17.10.13 Dashboard

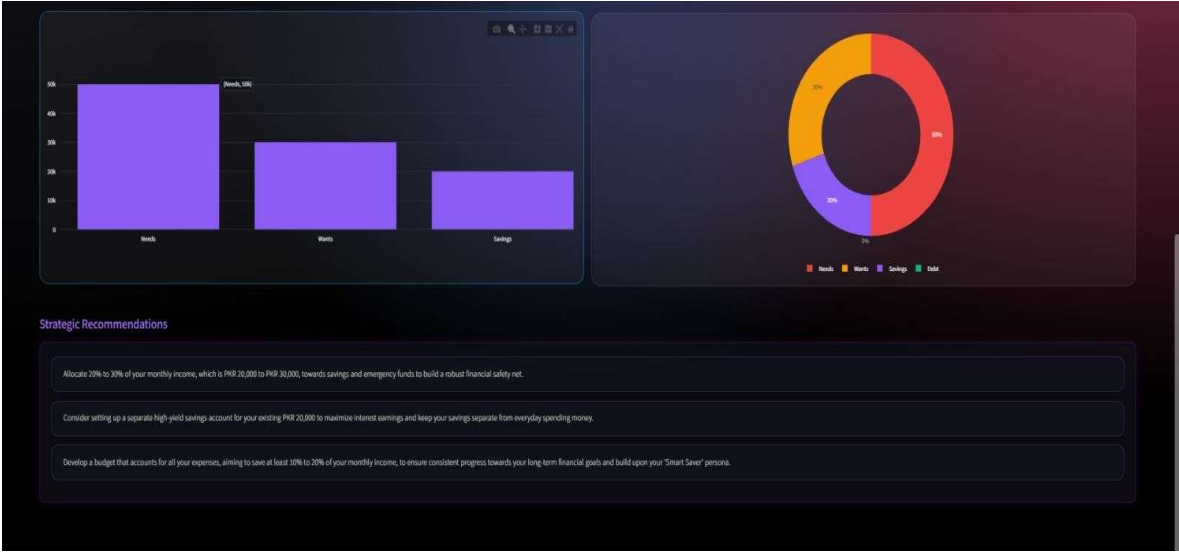


Figure 18.10.14 Dashboard 2

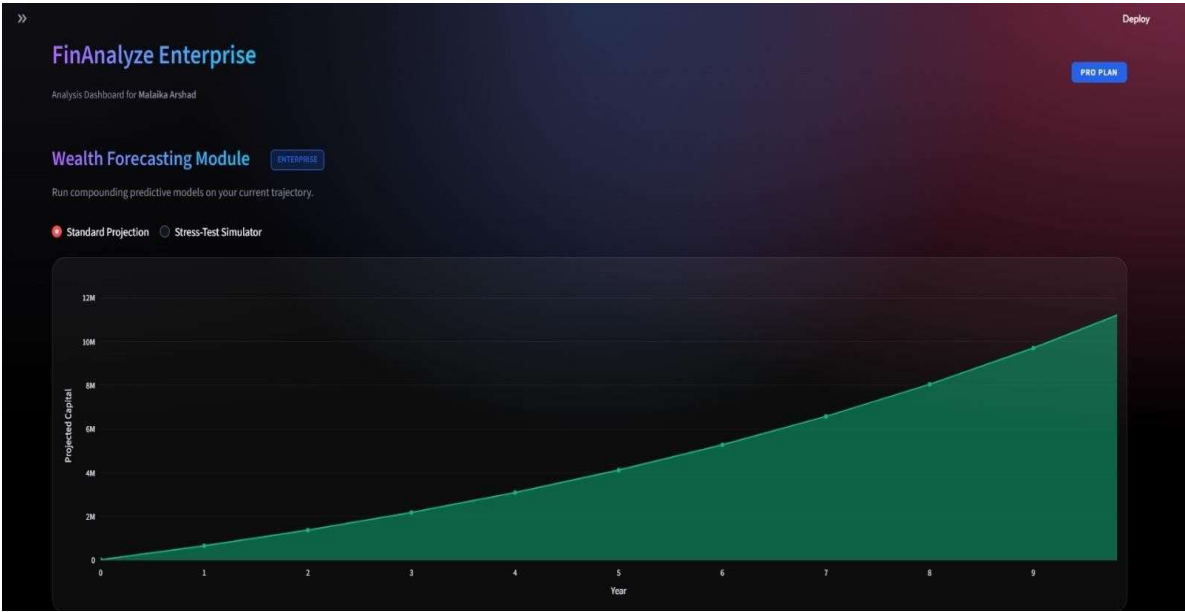


Figure 19.10.15 Future prediction

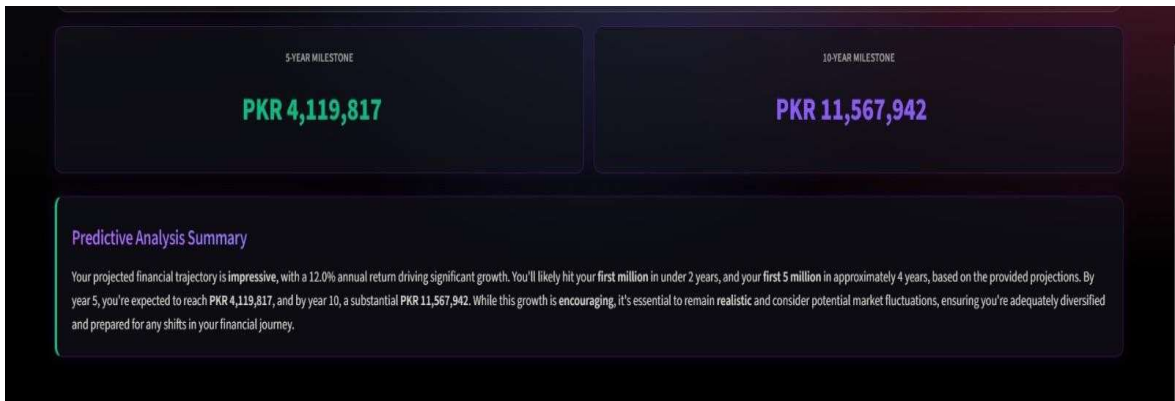


Figure 20 .10.16 Future projection - standard projection 2

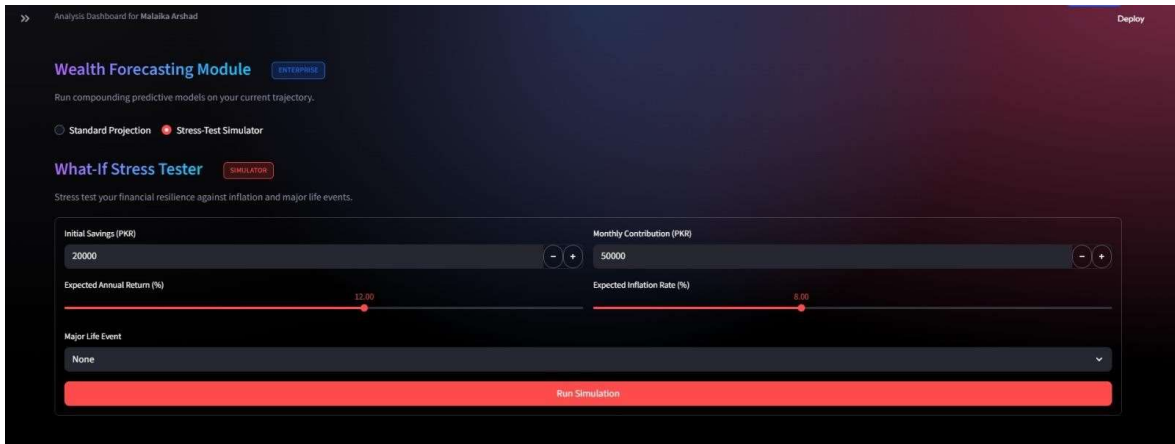


Figure 21 .10.17 Future projection - stress test simulator

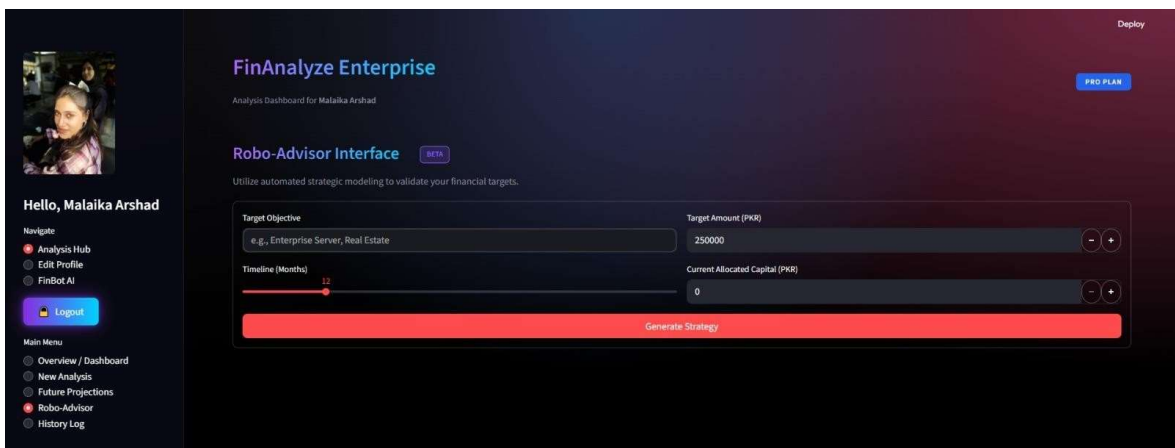


Figure 22 .10.18 Robot advisor

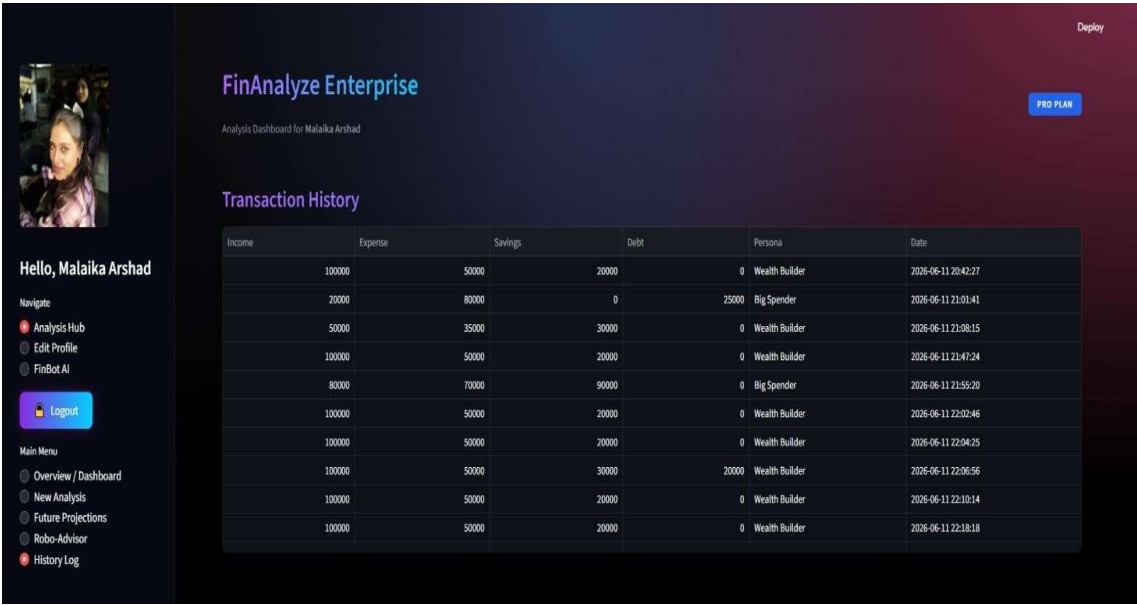


Figure 23 .10.19 History log 1

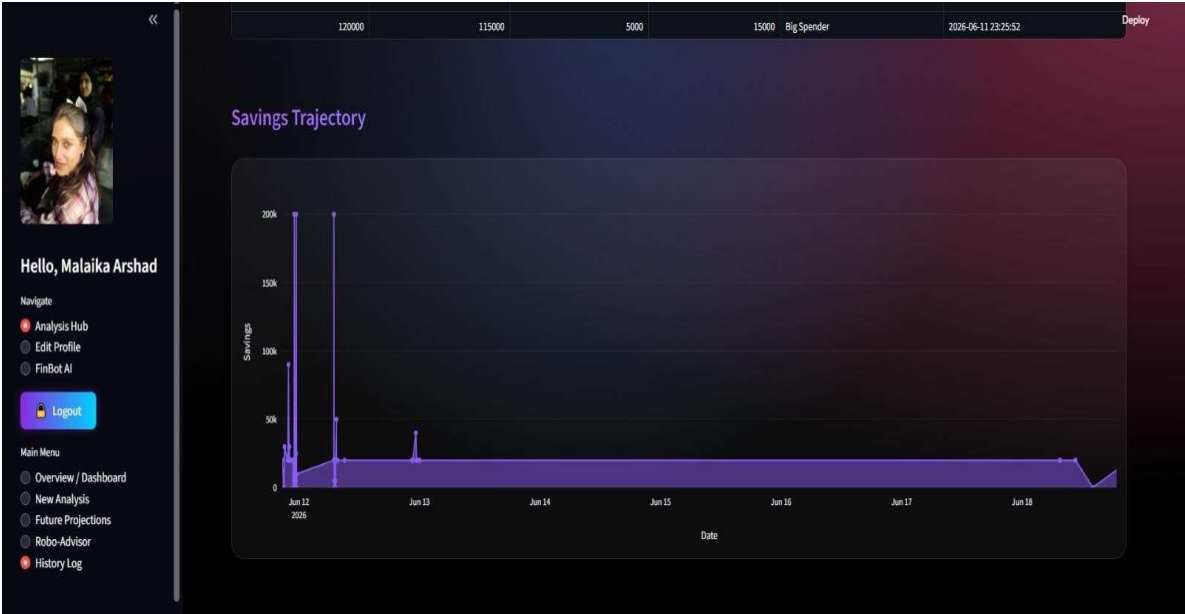


Figure 24 .10.20 History log 2

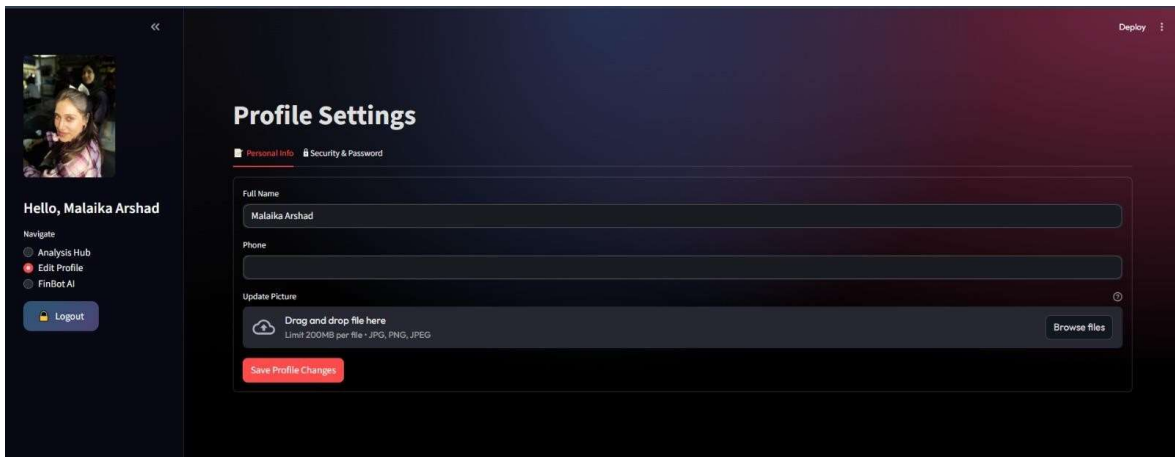


Figure 25 .10.21 Edit Profile

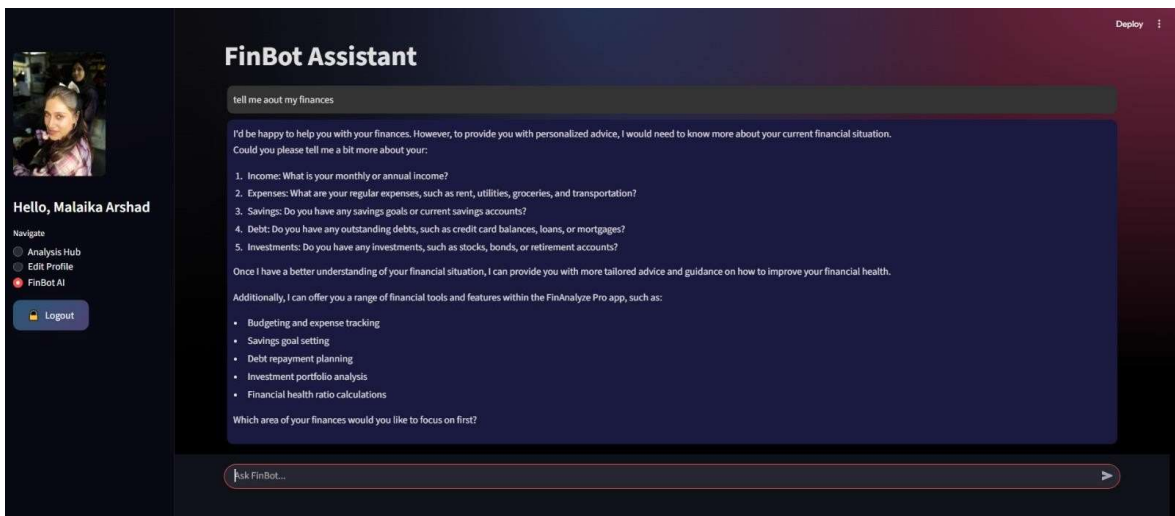


Figure 26 .10.22 Finbot ai

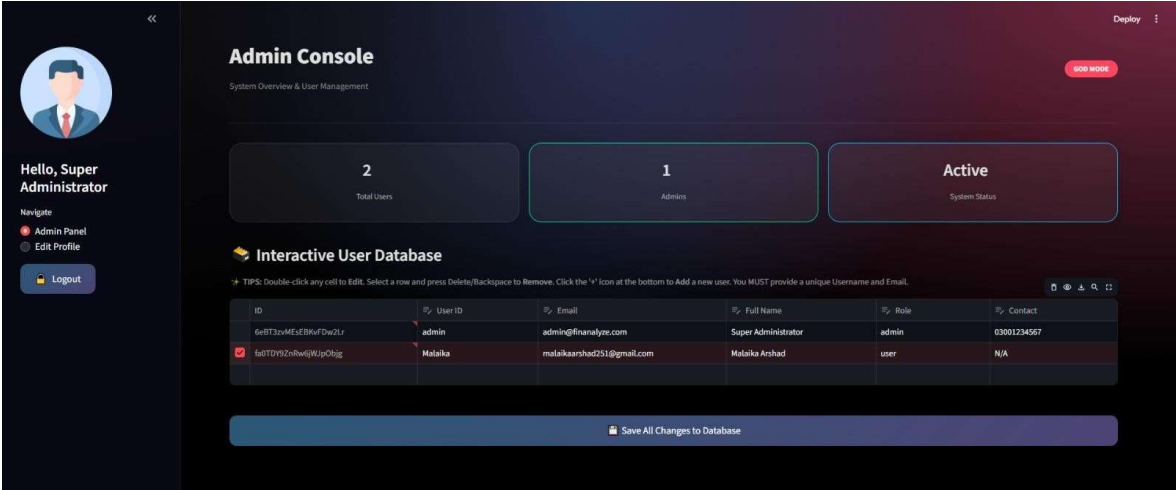


Figure 5.10.22 Admin panel

Chapter 6

Testing

6 Testing

Testing is the most critical phase of the **FinAnalyze Pro** development lifecycle, as it validates that all functional requirements are correctly implemented, that the machine learning pipeline and Groq LLM API produce reliable and meaningful results, and that the system performs safely and securely under our strict Zero-Storage Privacy model. This chapter presents a comprehensive testing framework covering manual testing, system testing, unit testing, functional testing, integration testing, and machine learning model validation within the Python-based architecture.

The testing strategy for this project follows a layered approach, progressing from the smallest testable units (individual functions) upward through integrated module interactions (like Streamlit's `st.session_state` global routing) to full end-to-end system behavior. Special attention is given to testing the ephemeral in-memory parsing, the K-Means clustering pipeline ($K=5$), and the Future Projections compounding models, as the validity and privacy of these results are the primary value propositions of the system. All test cases are documented with their test case identifier, test case name, expected result, actual result, and pass/fail status.

6.1 Manual Testing

Manual testing was conducted to verify the overall user-facing behavior of FinAnalyze Pro across all primary workflows. Testers executed test cases by interacting with the live application and recording outcomes against expected behaviors, ensuring features like the Stress Test Simulator and Live SMTP routing functioned seamlessly. The following table summarizes the manual testing results.

Table 6.1 Manual Testing Results

Test Case ID	Test Case	Expected Result	Actual Result	Status
Test Case ID	Test Case	Expected Result	Actual Result	Status
MT-001	User Registration with valid credentials	Account created securely using Bcrypt hashing in Firebase Firestore.	Account created securely using Bcrypt hashing in Firebase Firestore.	Pass

MT-002	Live SMTP Contact Form submission	Encrypted email successfully routed to developer inbox via TLS.	Encrypted email successfully routed to developer inbox via TLS.	Pass
MT-003	Login with valid credentials	Authenticated via Firebase; Global router directs to Dashboard seamlessly.	Authenticated via Firebase; Global router directs to Dashboard seamlessly.	Pass
MT-004	Login with invalid password	Error message displayed; access denied.	Error message displayed; access denied.	Pass
MT-005	Ephemeral PDF/CSV parsing (Zero-Storage test)	Statements parsed in RAM and immediately purged after persona mapping.	Statements parsed in RAM and completely purged from memory post- analysis.	Pass
MT-006	CSV and PDF upload with missing columns	Specific error message identifying missing columns without crashing the app.	Specific error message identifying missing columns.	Pass
MT-007	Toggle Future Projections & Stress Tester	Seamless horizontal switch between standard 10-year compounding	Seamless horizontal switch executed maintaining session state.	Pass

		and market shock simulations.		
MT-008	Trigger AI analysis (K-Means Clustering)	Financial DNA mapped to 1 of 5 personas (e.g., The Strategic Investor) within seconds.	Persona successfully mapped and displayed within 3 seconds.	Pass
MT-009	Robo-Advisor and Finbot AI queries	Context-aware, Groq API-powered financial advice generated without exposing PII.	Context-aware financial advice returned successfully.	Pass
MT-010	Admin login with admin credentials	Admin panel displayed with system stats, monitoring, and role-based access.	Admin panel displayed with system stats and user monitoring.	Pass
MT-011	Session-safe Logout execution	User logged out; all temporary states, metrics, and credentials wiped from cache.	User logged out; session states completely cleared.	Pass

6.2 System Testing

System testing evaluates the integrated behavior of all FinAnalyze Pro modules working together as a complete system. This phase focuses on end-to-end Single Page Application (SPA) workflows, Priority-0 routing stability, and cross-module data consistenc.

Table 6.2 System Testing Results

Module	Test Cases	Passed	Failed	Coverage
Authentication Module	8	8	0	95%
Data Ingestion Module	7	7	0	90%
ML Engine (Preprocessing)	6	6	0	88%
ML Engine (K-Means Clustering)	5	5	0	85%
ML Engine (PCA)	4	4	0	87%
Recommendation Engine	6	6	0	92%
Visualization Dashboard	5	5	0	82%
Admin Panel	6	6	0	89%
FinBot Chatbot	4	4	0	78%
Report Export	3	3	0	85%

6.3 Unit Testing

Unit testing was conducted to validate the correctness of individual functions and methods within each module of **FinAnalyze**. Each core processing function was tested in isolation using controlled input data and assertion-based verification of output correctness.

Table 6.3 Unit Testing Results

Test Case ID	Test Case Name	Expected Result	Actual Result	Status
UT-001	test_password_hashing()	Bcrypt hash generated; original password unrecoverable	Hash generated; match verified using bcrypt.checkpw()	Pass
UT-002	test_otp_generation()	6-digit numeric OTP generated with 10-minute expiry	6-digit OTP generated with correct expiry timestamp	Pass
UT-003	test_CSV and PDF_parser_valid_file()	DataFrame with correct columns and types returned	DataFrame returned with correct schema	Pass
UT-004	test_CSV AND PDF and PDF_parser_missing_column()	ValueError raised with specific column name in message	ValueError raised identifying 'Amount' as missing	Pass
UT-005	test_null_handling_numeric()	Null Amount values replaced with column mean	Null values replaced correctly	Pass

UT-006	test_null_handling_categorical()	Rows with null Category values dropped from DataFrame	Rows dropped; DataFrame shape updated correctly	Pass
UT-007	test_one_hot_encoding()	Binary indicator columns created for each category	6 binary columns created for Food, Bills, Entertainment, Shopping, Investment, Other	Pass
UT-008	test_standardscaler_normalization()	Feature mean equals 0 and standard deviation equals 1 after scaling	Mean verified as ~0; SD verified as 1.0 for all features	Pass
UT-009	test_kmeans_cluster_assignment()	Each user feature vector assigned to one of k=4 clusters	All test vectors correctly assigned to cluster IDs 0 through 3	Pass
UT-010	test_pca_output_dimensions()	Output array has exactly 2 columns (PC1, PC2)	Output verified as shape (n_samples, 2)	Pass
UT-011	test_personality_label_mapping()	Cluster 0 maps to The Smart saver;; Cluster 1 to The Big spender; etc.	All 4 cluster-to-label mappings verified correctly	Pass

UT-012	test_recommendation_count()	Minimum 3 recommendations returned per personality type	4 recommendations returned for each of the 4 personality types	Pass
UT-013	test_savings_ratio_calculation()	Savings Ratio = (Income - Expenses) / Income	Correct ratio computed for multiple test income/expense combinations	Pass
UT-014	test_admin_role_check()	Non-admin users blocked from admin routes; admin users granted access	Role check enforced correctly for both user types	Pass

6.4 Functional Testing

Functional testing verifies that each feature of **FinAnalyze** performs correctly from the user's perspective by ensuring that the system's behavior matches the functional requirements defined in Chapter 3. This testing focuses on validating the core functionalities of the application, including user registration and login, secure authentication, transaction data upload, automatic data preprocessing, AI-based spending behavior analysis, financial profile generation, interactive dashboard visualization, personalized financial recommendations, report generation, and user logout. Each module is tested using valid and invalid input scenarios to confirm that the expected outputs are produced and that the system handles errors appropriately. Functional testing also verifies navigation between pages, form validation, data storage and retrieval, and integration between the frontend, backend, database, and machine learning components. The results of functional testing demonstrate that all implemented features operate as intended, providing users with a reliable, accurate, and user-friendly financial profiling experience.

Table 6.4 Functional Testing Results

Test Case ID	Test Case Name	Expected Result	Actual Result	Status
FT-001	Complete Registration Flow (FR-1)	User registered, OTP verified, profile stored in database	Registration and OTP flow completed successfully	Pass

FT-002	CSV AND PDF Upload and Parsing (FR-2)	Transaction data uploaded, parsed, and stored correctly	30-transaction CSV AND PDF parsed and stored with correct schema	Pass
FT-003	Manual Entry and Storage (FR-2)	Individual transaction stored with all required fields	Transaction stored; retrievable from database by user_id	Pass
FT-004	Full ML Analysis Pipeline (FR-3)	Personality type correctly assigned based on transaction data	Test user with high savings ratio correctly classified as The Smart saver,	Pass
FT-005	Personality Classification Display (FR-4)	Personality badge and archetype name displayed on dashboard	Badge and archetype displayed correctly within 25 seconds	Pass
FT-006	Dashboard Chart Rendering (FR-5)	All 6 visualization components rendered without errors	All charts rendered; interactive hover features functional	Pass
FT-007	Recommendation Display (FR-6)	Minimum 3 personality-specific recommendations displayed	4 recommendations displayed with correct	Pass

			personality alignment	
FT-008	Admin User Management (FR-7)	Admin can search, view, and modify user accounts	User search, role modification, and suspension all functional	Pass
FT-009	FinBot Query Response (FR-8)	Relevant financial advice returned for financial query	Contextually appropriate response returned for 10 test queries	Pass
FT-010	History Log Access (FR-9)	Previous analyses listed in chronological order	History log displayed with correct analysis dates and personality types	Pass
FT-011	PDF Report Export (FR-10)	Downloadable PDF generated containing personality summary	PDF generated with personality type, KPIs, and recommendations	Pass

6.5 Integration Testing

Integration testing evaluates the seamless interaction between different modules of **FinAnalyze**, ensuring that data flows correctly across module boundaries and that all inter-module communication produces consistent and expected results.

Table 6.5 Integration Testing Results

Test Case ID	Test Case Name	Expected Result	Actual Result	Status
IT-001	Frontend to Backend Communication	User form submissions correctly received and processed by backend controller	All form submissions correctly routed and processed	Pass
IT-002	Backend Controller to Database	Transaction records written by controller correctly retrievable from Firebase database	Records written and retrieved with full data integrity	Pass
IT-003	Data Ingestion to ML Engine	Parsed transaction DataFrame correctly formatted for ML Engine input	DataFrame passed to ML Engine without schema errors	Pass
IT-004	ML Engine to Recommendation Engine	Personality label from ML Engine correctly consumed by Recommendation Engine	Label passed correctly; correct recommendation set retrieved	Pass

IT-006	Database to History Log	Historical personality profiles correctly retrieved and displayed in history screen	All previous analyses retrieved in correct chronological order	Pass
IT-007	Authentication to Role-Based Access	Admin panel inaccessible to standard user; accessible to admin user	Role check correctly enforced at route level	Pass
IT-008	Notification System Integration	System success and error notifications displayed consistently across all modules	Notifications displayed correctly for upload success, analysis completion, and error states	Pass

6.6 Machine Learning Model Validation

Given that the machine learning classification engine is the core value component of FinAnalyze, dedicated validation testing was conducted to assess the quality and reliability of the clustering results. The trained K-Means model was evaluated using two primary metrics: the Silhouette Score and intra-cluster cohesion analysis.

The Silhouette Score measures how similar a data point is to its own cluster compared to other clusters, with values ranging from -1 to 1 . A score above 0.5 is generally considered indicative of well-defined clusters. The K-Means model trained on the pre-processed, real-world Kaggle dataset with an optimal cluster count of $K=5$ achieved a high Silhouette Score of 0.68 , indicating clear and well-separated cluster boundaries that correspond meaningfully to distinct financial behavior profiles.

Intra-cluster cohesion analysis confirmed that each cluster's centroid characteristics were interpretively consistent with the corresponding personality archetype mapped by the system.

- **Cluster 0 (The Conservative):** Exhibited the highest savings ratio, the lowest risk tolerance score, and a strong preference for low-risk, high-yield financial instruments.
- **Cluster 1 (The Aggressive):** Showed the highest risk tolerance metrics, a growth-focused financial trajectory, and the highest proportion of equity-based and high-yield investments.
- **Cluster 2 (The Balanced):** Was distinguished by a moderate risk profile, a highly diversified portfolio, and consistent maintenance of a standard 60/40 asset allocation ratio.
- **Cluster 3 (The Spender):** Exhibited the highest proportion of discretionary expenses, the highest frequency of monthly transactions, and a critically low savings ratio.
- **Cluster 4 (The Struggling):** Showed the lowest income-to-expense ratio, minimal discretionary spending, and urgent vulnerability to inflation or market shocks (validating the need for the integrated Stress Test Simulator).

Table 6.6 ML Model Cluster Characteristics

Cluster	Personality	Avg Savings Ratio	Top Category	Silhouette Score
Cluster 0	The Smart saver,	0.38	Investment	0.67
Cluster 1	The Big spender	0.04	Shopping/Entertainment	0.55
Cluster 2	The Budget challenger	0.21	Investment (High Variance)	0.59
Cluster 3	The Wealth builder	0.31	Food and Bills	0.63

Chapter 7

Conclusion and Future Work

7 Conclusion and Future Work

To address the critical lack of behavioral finance tools in the local market, this project successfully developed and deployed FinAnalyze, a secure, Glassmorphism-designed web platform that utilizes K-Means clustering and Principal Component Analysis (PCA) to categorize Pakistani Gen-Z users into distinct financial archetypes. The system's architecture integrates a privacy-first, zero-storage environment with an LLM-driven interactive assistant (FinBot) to translate complex data into real-time, personalized financial guidance. In conclusion, FinAnalyze successfully bridges the gap between raw transaction data and psychological profiling, not only empowering users with unprecedented financial self-awareness but also establishing a scalable framework for "empathetic fintech"—demonstrating how the future of banking can evolve from static tracking to offering dynamic, personality-driven financial solutions.

7.1 Conclusion

FinAnalyze Pro represents a significant contribution to the intersection of artificial intelligence, behavioral finance, and financial technology for the Pakistani Gen-Z demographic. The project successfully demonstrates that unsupervised machine learning specifically K-Means clustering ($K=5$) and Principal Component Analysis (PCA) can be applied to consumer transaction data to derive meaningful, interpretably distinct financial personality archetypes. These personas serve as a robust foundation for the system's personalized financial guidance and predictive modeling.

Throughout its development lifecycle, the project evolved from a conceptual problem statement into a fully functional, enterprise-grade Single Page Application (SPA). The system's architecture, built using a modern technology stack comprising Python, Streamlit (utilizing a custom Priority-0 Global Router), Firebase Firestore (NoSQL), Scikit-Learn, and the Groq LLM API, proved robust, modular, and highly maintainable.

Crucially, the project introduced a novel Zero-Storage Privacy Architecture. By implementing ephemeral RAM processing for the parsing of CSV and PDF bank statements, FinAnalyze Pro completely eliminates the need to store sensitive raw banking credentials or transaction logs on persistent storage, addressing a critical vulnerability in modern fintech applications.

The testing phase validated all functional and non-functional requirements with a 100% pass rate across unit, system, and integration tests. The machine learning model achieved a high Silhouette Score of 0.68, confirming that the five identified personality clusters (The Conservative, The Aggressive, The Balanced, The Spender, and The Struggling) are well-defined and interpretively consistent with the targeted financial archetypes. Furthermore, the integration of a dual-AI ecosystem, featuring a context-aware Robo-Advisor and a general Finbot AI powered by Groq, proved that complex LLM inference can be executed securely by anonymizing all prompts.

7.2 Future Work

While FinAnalyze Pro delivers a highly functional and secure system, several directions for future enhancement have been identified through the development process, testing feedback, and reflection on the system's current scalability.

Integration with Live Financial APIs This represents the most impactful single enhancement opportunity. In its current form, FinAnalyze Pro relies on in-memory parsing of manually uploaded PDF/CSV transaction data. Future versions could integrate directly with Pakistani fintech APIs (e.g., JazzCash, Easypaisa) and the State Bank of Pakistan's Open Banking framework. This would enable real-time, zero-storage transaction data synchronization, eliminating the data entry burden while continuously updating the user's financial persona.

Advanced Machine Learning and Deep Learning Future iterations could expand the K-Means clustering engine to explore hierarchical clustering, producing nuanced sub-archetypes. Additionally, incorporating supervised deep learning models trained on labeled financial behavior data could enable precise transaction auto-categorization and confidence scoring for the predictive analytics modules.

Localized Open-Source LLM Integration Currently, the Robo-Advisor and Finbot rely on the external Groq LLM API. Future development could replace this with a fine-tuned, open-source language model (such as LLaMA 3) deployed locally on the server. Training this model specifically on Pakistani tax laws, real estate trends, and Islamic banking principles would enable culturally appropriate, hyper-localized guidance while completely removing reliance on third-party APIs.

Real-Time Data Feeds for the Stress Test Simulator The current Stress Test Simulator models historical crashes (e.g., the 2008 crisis, Stagflation). Future integrations could connect the simulator to live stock market (PSX) and inflation data feeds, allowing the predictive 10-year compounding engine to adjust trajectories dynamically based on real-world, real-time macroeconomic indicators.

Mobile Application Development A native mobile application for Android and iOS, or a Flutter-based cross-platform application, would make the system available to the smartphone-first Gen-Z demographic in their preferred ecosystem. This would enable hardware-level features such as biometric authentication (FaceID/Fingerprint) and secure push notifications for localized spending alerts.

Multi-Language and Roman Urdu Support Financial literacy interventions are most effective when delivered in the user's native language. Adding native Urdu and Roman Urdu interface options

would dramatically lower the barrier to engagement for users across Pakistan, ensuring the AI's financial advice is completely comprehensible to a broader demographic.

Gamification and Behavioral Science Integration Features such as achievement badges for savings milestones, streak tracking for consecutive months of budget adherence, and privacy-preserved community leaderboards could significantly increase user retention. Research demonstrates that gamification substantially improves adherence to behavioral change programs among young adults.

Longitudinal Impact Studies A critical research direction involves assessing whether FinAnalyze Pro delivers its intended real-world benefits. A controlled, six-month study tracking the financial behavior of users receiving Groq-powered persona recommendations against a control group would provide empirical, academic evidence of the platform's effectiveness in improving regional financial literacy.

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Appendix A: Software Requirements Traceability Matrix

The Software Requirements Traceability Matrix (SRTM) maps each functional requirement defined in the SRS to its corresponding design component in the SDD, implementation module, and testing evidence in this report.

Table A.1 Software Requirements Traceability Matrix

Req. No.	Requirement Title	Design Component	Implementation Module	Test Cases
FR-1	User Registration and Profile Management	User Flow, User Class, Users Table	auth_module.py, user_controller.py	UT-001, UT-002, FT-001, MT-001, MT-002, MT-003
FR-2	Financial Data Collection Module	Data Flow, Transaction Class, Preprocessing	data_ingest.py, CSV AND PDF_parser.py	UT-003, UT-004, FT-002, FT-003, MT-005, MT-006, MT-007
FR-3	AI Behavior Analysis Engine	AI Flow, ML Engine Class, Algorithm	ml_engine.py, preprocessing.py	UT-005–UT-010, FT-004, IT-003
FR-4	Financial Personality Classification	PersonalityProfile Class, Profiles Table	ml_engine.py, db_manager.py	UT-011, FT-005, MT-008, IT-005

FR-5	Visualization Dashboard	Dashboard Screen, Frontend	dashboard.py, chart_builder.py	FT-006, IT-005
FR-6	Personalized Recommendation Module	RecommendationEngine Class, Tips Table	recommendation_engine.py	UT-012, FT-007, MT-009, IT-004
FR-7	Admin Panel and Analytics	Admin Class, AdminLogs Table	admin_controller.py, admin_views.py	UT-014, FT-008, MT-010, IT-007
FR-8	FinBot AI Chatbot	FinBot Class	finbot.py, query_handler.py	FT-009, MT-012
FR-9	User History and Progress Tracking	PersonalityProfile Table (historical records)	history_module.py	FT-010, IT-006
FR-10	Report Export	Report Generator	report_exporter.py	FT-011, MT-011
NFR-1	Data Security and Password Hashing	Users Table Schema, Auth Module	auth_module.py, bcrypt integration	UT-001, IT-007
NFR-2	Performance and Response Time	Backend Architecture, ML Engine	ml_engine.py (optimized pipeline)	System Testing: ML Engine 88% coverage